

Appendix N

Supporting information for case studies

In the previous editions, we provided Fortran code to carry out the simulations that were discussed in the Case Studies. In the current edition, we provide some Jupiter-Notebooks instead.

Most of the old case studies are now presented as Examples in the new edition. For those that are interested in running the FORTRAN code, we list all the case studies from the second edition here.

N.1 Equation of state of the Lennard-Jones fluid-I

Case Study 1 (Equation of state of the Lennard-Jones fluid-I). One important application of molecular simulation is computing the phase diagram of a given model system. In several chapters, we discuss some of the techniques that have been developed to study phase transitions, including direct-coexistence simulations. However, direct-coexistence simulations may suffer from hysteresis, which may make them less suited for locating phase transitions. In the present Case Study, we illustrate some of the problems that occur when we use standard Monte Carlo simulation to determine a phase diagram. As an example, we focus on the vapor-liquid curve of the Lennard-Jones fluid. Of course, as was already mentioned in section 3.3.2.2, the phase behavior is quite sensitive to the detailed form of the intermolecular potential that is used. In the Case Study, we approximate the full Lennard-Jones potential as follows:

$$u(r) = \begin{cases} u_{\text{LJ}}(r) & r \leq r_c \\ 0 & r > r_c, \end{cases}$$

where the cutoff radius r_c is set to half the box length. The contribution of the particles beyond this cutoff is estimated with the usual tail corrections; that is, for the energy

$$u^{\text{tail}} = \frac{8}{3}\pi\rho \left[\frac{1}{3} \left(\frac{1}{r_c} \right)^9 - \left(\frac{1}{r_c} \right)^3 \right]$$

and for the pressure

$$p^{\text{tail}} = \frac{16}{3}\pi\rho^2 \left[\frac{2}{3} \left(\frac{1}{r_c} \right)^9 - \left(\frac{1}{r_c} \right)^3 \right].$$

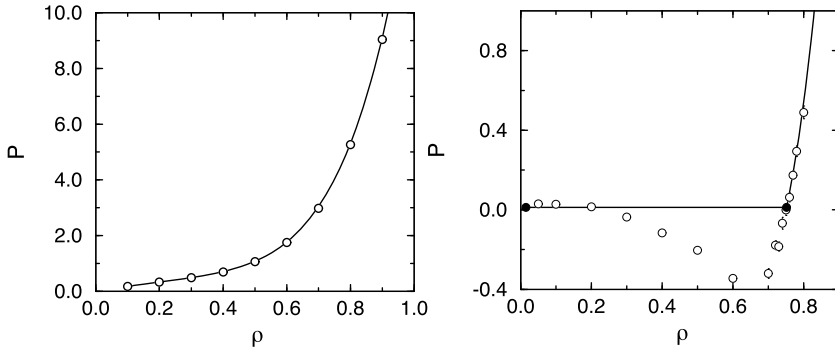


FIGURE N.1 Equation of state of the Lennard-Jones fluid. (Left) Isotherm at $T = 2.0$. (Right) Isotherm below the critical temperature ($T = 0.9$); the horizontal line is the saturated vapor pressure, and the filled circles indicate the densities of the coexisting vapor and liquid phases. The solid curve represents the equation of state of Johnson et al. [83], and the circles are the results of the simulations ($N = 500$). The horizontal line in the right-hand figure corresponds to the coexistence pressure obtained by applying the Maxwell construction to the Johnson equation of state. The Maxwell construction exploits the fact that the equality of the chemical potentials of the coexisting phases implies that $P_{\text{coex}} = \int_{V_{\text{liq}}}^{V_{\text{vap}}} dV P(V)/(V_{\text{vap}} - V_{\text{liq}})$. The statistical errors in the numerical data are smaller than the symbol sizes.

The equation of state of the Lennard-Jones fluid has been investigated by many groups using Molecular Dynamics or Monte Carlo simulations starting with the work of Wood and Parker [69]. The first systematic study of the equation of state of the Lennard-Jones fluid was reported by Verlet [14]. Subsequently, many more studies have been published. In 1979, the data available at that time were compiled by Nicolas et al. [82] into an accurate equation of state. This equation has been refitted by Johnson et al. [83] using the best data then available. In the present study, we compare our numerical results with the equation of state by Johnson et al.

We performed several simulations using Algorithms 1 and 2. During the simulations we determined the energy per particle and the pressure. The pressure was calculated using the virial, W

$$P = \frac{\rho}{\beta} + \frac{W}{V}, \quad (\text{N.1.1})$$

where the virial is defined by

$$W = \frac{1}{3} \sum_i \sum_{j>i} \mathbf{f}(\mathbf{r}_{ij}) \cdot \mathbf{r}_{ij}, \quad (\text{N.1.2})$$

where $\mathbf{f}(\mathbf{r}_{ij})$ is the intermolecular force. Fig. N.1 (left) compares the pressure as obtained from a simulation above the critical temperature with the equation of state of Johnson et al. [83]. The agreement is excellent (as is to be expected).

Fig. N.1 (right) shows a typical isotherm below the critical temperature. If we cool the system below the critical temperature, we should expect to observe

vapor-liquid coexistence. However, standard Monte Carlo or Molecular Dynamics simulations of small model systems are not suited to study the coexistence between two phases. Using the Johnson equation of state, we predict how the pressure of a macroscopic Lennard-Jones system would behave in the two-phase region (see Fig. N.1). For densities inside the coexistence region, the pressure is expected to be constant and equal to the saturated vapor pressure. If we now perform a Monte Carlo simulation of a finite system (500 LJ particles), we find that the computed pressure is not at all constant in the coexistence region (see Fig. N.1). In fact we observe that, over a wide density range, the simulated system is metastable and may even have a negative pressure. The reason is that, in a finite system, a relatively important free-energy cost is associated with the creation of a liquid-vapor interface. So much so that, for sufficiently small systems, it is favorable for the system not to phase separate at all [85]. Clearly, these problems will be most severe for small systems and in cases where the interfacial free energy is large. For this reason, standard NVT -simulations are not recommended to determine the vapor-liquid coexistence curve or, for that matter, any strong first-order phase transition in small systems.

To determine the liquid-vapor coexistence curve, we should determine the equation of state for a large number of state points outside the coexistence region. These data can then be fitted to an analytical equation of state. With this equation of state, we can determine the vapor-liquid curve (this is the procedure used by Nicolas et al. [82] and Johnson et al. [83]).

Of course, if we simulate a system consisting of a sufficiently large number of particles, it is possible to simulate a liquid phase in coexistence with its vapor. However, such simulations are time-consuming, because Monte Carlo moves are not very efficient at equilibrating the pressure in a two-phase system.

N.2 Importance of detailed balance

Case Study 2 (Importance of detailed balance). Monte Carlo simulations aim to sample points in configuration space according to their Boltzmann weights. If the system is ergodic, imposing detailed balance is sufficient, but not necessary to ensure Boltzmann sampling. In other words: some sampling schemes that violate detailed balance may nevertheless lead to correct Boltzmann sampling (see Section 13.4.4). However, one should avoid non-detailed-balance schemes, unless one can prove that they satisfy the more general *balance* condition. In the present Example, we show that systematic errors result when using a scheme that does not satisfy the *balance* condition.

Consider an ordinary N, V, T move; a new trial position is generated by giving a randomly selected particle, say i , a random displacement:

$$x_n(i) = x_o(i) + \Delta_x(\mathcal{R} - 0.5),$$

where Δ_x is twice the maximum displacement. We now make a seemingly small change in the algorithm and generate a new position using

$$x_n(i) = x_o(i) + \Delta_x(\mathcal{R} - 0.0) \quad \text{wrong!}$$

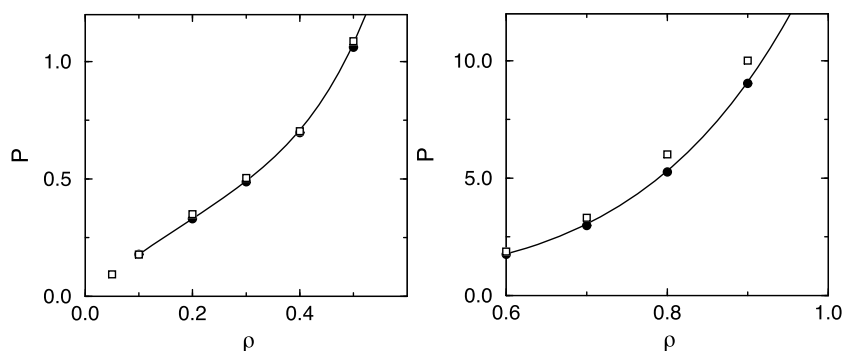


FIGURE N.2 Equation of state of the Lennard-Jones fluid ($T = 2.0$); comparison of a displacement scheme that obeys detailed balance (circles) and one that does not (squares). Both simulations have been performed with 500 particles. The solid curve is the equation of state of Johnson et al. [83]. The figure at the left corresponds to the low-pressure regime. The high-pressure regime is shown in the right-hand figure.

i.e., we give the particles only a *positive* displacement. With such a move, detailed balance is violated, since the reverse move — putting the particle back at x_o — is not possible.¹

For the Lennard-Jones fluid, we can use the program of Case Study 1 to compare the two sampling schemes. The results of these simulations are shown in Fig. N.2. At first sight, the results of the incorrect scheme look reasonable; in fact, at low densities, the results of the two schemes do not show significant differences. But at high densities, the incorrect scheme overestimates the pressure. This overestimate of the pressure is a systematic error: it does not disappear when we perform longer simulations.

The above example illustrates something important: one cannot decide if an algorithm is correct on the basis of the fact that the results *look* reasonable. Simulation results must always be tested on cases where we *know* the correct answer: either a known (and trusted) numerical result, but preferably an exact result that may be known in some limiting case (dilute vapor, dense solid, etc.).

Usually, one does not know *a priori* the size of the optimal maximum displacement in a Monte Carlo simulation. By doing short test runs, we can explore what maximum displacement achieves the best sampling for a given amount of simulation time. However, the maximum step-size should not be changed during a production run, because then one would violate detailed balance [90]: if from one move to the next, the maximum displacement is decreased, then the *a priori* probability for a particle to return to its previous position could be zero, which violates microscopic reversibility.

¹ In Event-Chain MC (section 13.4.4) we will see examples of forward-only algorithms that have been designed to satisfy balance.

N.3 Why count the old configuration again?

Case Study 3 (Why count the old configuration again?). A somewhat counter-intuitive feature of the Metropolis sampling scheme is that, if a trial move is rejected, we should once again count the contributions of the old configuration to the average that we are computing (see acceptance rule (3.2.12)). The aim of this Case Study is to show that this recounting is really essential. In the Metropolis scheme, the acceptance rule for a move from o to n is

$$\begin{aligned} \text{acc}(o \rightarrow n) &= \exp\{-\beta[\mathcal{U}(n) - \mathcal{U}(o)]\} & \mathcal{U}(n) \geq \mathcal{U}(o) \\ &= 1 & \mathcal{U}(n) < \mathcal{U}(o). \end{aligned}$$

These acceptance rules lead to a transition probability

$$\begin{aligned} \pi(o \rightarrow n) &= \exp\{-\beta[\mathcal{U}(n) - \mathcal{U}(o)]\} & \mathcal{U}(n) \geq \mathcal{U}(o) \\ &= 1 & \mathcal{U}(n) < \mathcal{U}(o). \end{aligned}$$

Note that this transition probability must be normalized:

$$\sum_n \pi(o \rightarrow n) = 1.$$

From this normalization, it follows that the probability that we accept the old configuration again is by definition

$$\pi(o \rightarrow o) = 1 - \sum_{n \neq o} \pi(o \rightarrow n).$$

This last equation implies that we should count the contribution of the old configuration again.

It is instructive to use the Lennard-Jones program from Case Study 1 to investigate numerically the error that is made when we only include accepted configurations in our averaging. In essence, this means that in Algorithm 2 we continue attempting to displace the selected particle until a trial move has been accepted.² In Fig. N.3 we compare the results of the correct scheme with those obtained by the scheme in which we continue to displace a particle until a move is accepted. Again the results look reasonable, but the figure shows that large, systematic errors are being made.

N.4 Static properties of the Lennard-Jones fluid

Case Study 4 (Static properties of the Lennard-Jones fluid). Let us illustrate the results of the previous sections with an example. Like in the section on Monte Carlo simulations we choose the Lennard-Jones fluid as our model system. We use

² It is easy to see that this approach leads to the wrong answer if we try to compute the average energy of a two-level system with energy levels E_0 and E_1 . If we include only accepted trial moves in our averaging, we would find that $\langle E \rangle = (E_0 + E_1)/2$, independent of temperature.

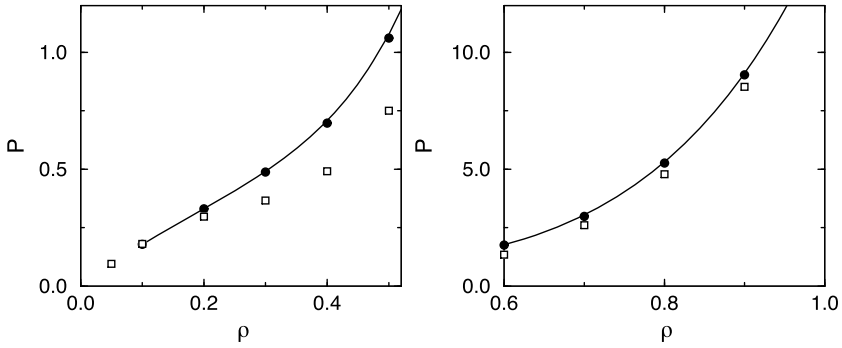


FIGURE N.3 Equation of state of the Lennard-Jones fluid ($T = 2.0$); comparison of a scheme in which particles are displaced until a move is accepted (squares) with the conventional scheme (circles). Both simulations have been performed with 108 particles. The solid curve is the equation of state of Johnson et al. [83]. The left figure is at low pressure, and the right one at high pressure.

a truncated and shifted potential (see also section 3.3.2.2):

$$u^{\text{tr-sh}}(r) = \begin{cases} u^{\text{lj}}(r) - u^{\text{lj}}(r_c) & r \leq r_c \\ 0 & r > r_c \end{cases}, \quad (\text{N.4.1})$$

where $u^{\text{lj}}(r)$ is the Lennard-Jones potential and for these simulations $r_c = 2.5\sigma$ is used.

During a simulation, we must check if the system has equilibrated or at least has reached a state that is stable on the timescale of the simulations. We then collect data about observable quantities, and at the end of the simulation, we compute averages and estimate the statistical errors. This Example illustrated how such a simulation might proceed.

At the beginning of the simulation, we prepare the system in a state that, of course, is not yet equilibrated. Here, we will assume that the particles were originally positioned on the lattice sites of a face-centered cubic crystal.

We initialize the velocities of the particles such that the initial kinetic energy corresponds to a temperature $T = 0.728$. The density is fixed at $\rho = 0.8442$, which is a typical liquid density, close to the triple (gas-liquid-solid) point of the Lennard-Jones fluid. When we start the MD simulation from this initial configuration, the potential energy will decrease, and because energy is conserved, the kinetic energy will go up—see Fig. N.4.

In Fig. N.4 shows the time evolution from the start of the simulation of the total energy, kinetic energy, and potential energy. Note that the total energy, whilst fluctuating slightly, does not drift. The kinetic and potential energies change strongly during the equilibration period, but after that, they oscillate around their equilibrium value. The figure suggests that for this (very small) system, equilibration is complete in less than 1000 time steps. However, longer equilibrations will be needed for larger systems, and for glassy systems, MD may not achieve equilibration at all.

Next, we consider error estimates. We used the method of Flyvbjerg and Petersen [148] to estimate the statistical error in the potential energy. We denote the

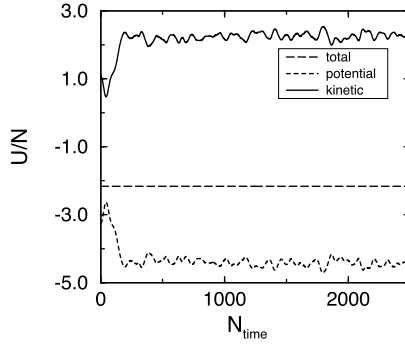


FIGURE N.4 Total, potential, and kinetic energy per particle U/N as a function of the number of time steps N_{time} .

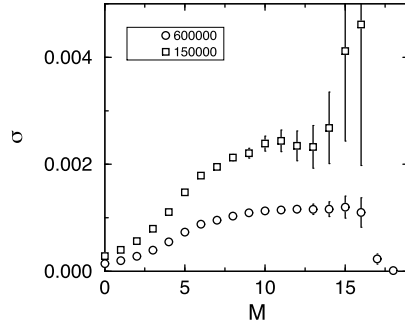


FIGURE N.5 The standard deviation σ in the potential energy as a function of the number of block operations M for a simulation of 150,000 and 600,000 time steps. This variance is calculated using Eq. (5.3.28).

number of blocking operations by M (see Fig. N.5). In this figure, the number of blocking operations by M , and from the plateau, we obtain an estimate of the standard deviation in our results.

The figure also shows the effect of increasing the total length of the simulation by a factor of 4; as expected, the statistical error in the potential energy decreases by a factor of 2.

We obtain the following results: for the potential energy $U = -4.4190 \pm 0.0012$ and for the kinetic energy $K = 2.2564 \pm 0.0012$, the latter corresponding to an average temperature of $T = 1.5043 \pm 0.0008$. For the pressure, we find 5.16 ± 0.02 .

In Fig. N.6, the radial distribution function is shown. To determine $g(r)$, we used Algorithm 8. This $g(r)$ shows the characteristics of a dense liquid. We can use the radial distribution function to calculate the energy and pressure. The potential energy per particle can be calculated from

$$\begin{aligned} U/N &= \frac{1}{2} \rho \int_0^\infty dr u(r) g(r) \\ &= 2\pi\rho \int_0^\infty dr r^2 u(r) g(r) \end{aligned} \quad (\text{N.4.2})$$

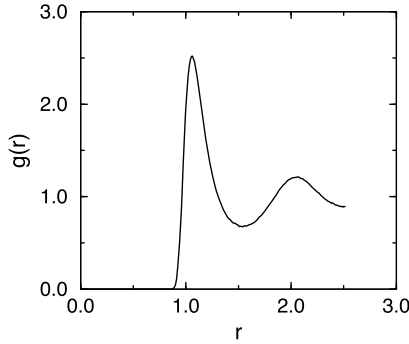


FIGURE N.6 Radial distribution function of a Lennard-Jones fluid close to the triple point: $T = 1.5043 \pm 0.0008$ and $\rho = 0.8442$.

and for the pressure from

$$\begin{aligned}
 P &= \rho k_B T - \frac{1}{3} \frac{1}{2} \rho^2 \int_0^\infty dr \frac{du(r)}{dr} r g(r) \\
 &= \rho k_B T - \frac{2}{3} \pi \rho^2 \int_0^\infty dr \frac{du(r)}{dr} r^3 g(r),
 \end{aligned} \tag{N.4.3}$$

where $u(r)$ is the pair potential.

Eqs. (N.4.2) and (N.4.3) can be used to check the consistency of the energy and pressure calculations and the determination of the radial distribution function. In our example, we obtained from the radial distribution function for the potential energy $U/N = -4.419$ and for the pressure $P = 5.181$, which is in good agreement with the direct calculation.

N.5 Dynamic properties of the Lennard-Jones fluid

Case Study 5 (Dynamic properties of the Lennard-Jones fluid). As an example of the numerical determination of a dynamical property, we consider the calculation of the self-diffusion coefficient. As shown in the previous section, the diffusion coefficient can be determined either from the mean-squared displacement, or from the velocity auto-correlation function (VACF). Here, we follow the approach of Algorithm 9 to compute these observables.

In Fig. N.7 the mean-squared displacement is shown as a function of the simulation time. From the mean-squared displacement, we can determine the diffusion using Eq. (5.2.13). This equation is however only valid in the limit $t \rightarrow \infty$. In practice, this means that we have to verify that we compute the mean-squared displacement in the long-time regime where all transient effects have died down, and only the linear dependence on t remains.

The [Green-Kubo \(GK\)](#) integral of the [Velocity autocorrelation function \(VACF\)](#) provides an alternative route to calculate the diffusion coefficient (Eq. (5.2.16)). Strictly speaking, the upper limit of the Green-Kubo integral corresponds to $t \rightarrow \infty$. In practice, we must stop at an earlier time t_c , because once the [VACF](#) has decayed into the noise, further integration increases the noise but not the signal. A rough

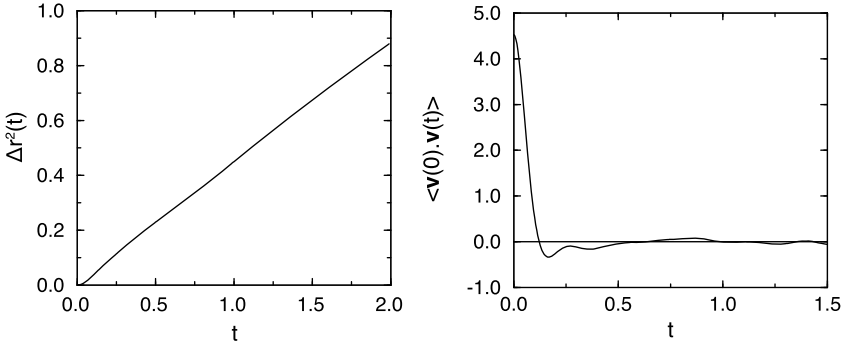


FIGURE N.7 (Left) Mean-squared displacement $\Delta r(t)^2$ as a function of the simulation time t . Note that for long times, $\Delta r(t)^2$ varies linearly with t . The slope is then given by $2dD$, where d is the dimensionality of the system and D is the self-diffusion coefficient. (Right) Velocity autocorrelation function $\langle \mathbf{v}(0) \cdot \mathbf{v}(t) \rangle$ as a function of the simulation time t .

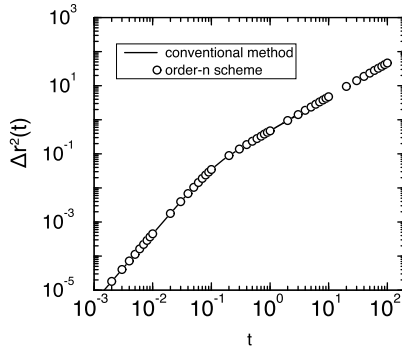


FIGURE N.8 Mean-squared displacement as a function of time for the Lennard-Jones fluid ($\rho = 0.844$, $N = 108$, and $T = 1.50$); comparison of the conventional method with the order- n scheme.

estimate of t_c can be obtained by determining the time when the integral seems to have reached a plateau value. Note that beyond t_c , we expect all transients in the mean-squared displacement to have died down.

N.6 Algorithms to calculate the mean-squared displacement

Case Study 6 (Algorithms to calculate the mean-squared displacement). In this case study, we compare the conventional (Algorithm 9), and the order- n methods (Algorithm 10) to determine the time-dependence of mean-squared displacement. The example considers a Lennard-Jones fluid.

In Fig. N.8 we compare the time-dependence of the mean-squared displacement computed with the conventional method and the one obtained with the order- n scheme. For the order- n scheme, the memory requirement grows logarithmically with time, but for the conventional scheme, it grows linearly. For the present simulation settings, the calculation using the conventional scheme could

not be extended beyond $t > 10$. For the same memory allocation the order- n scheme the calculation could be extended to much longer times with no difficulty: it only depends on the simulation time. In the present example, we stopped at $t = 200$. It is interesting to compare the accuracy of the two schemes. In the conventional scheme, the velocities of the particles at the current time step are used to update the mean-squared displacement of all time intervals. In the order- n scheme, the current time step is only used to update the lowest-order array of $\mathbf{v}_{\text{sum}}$ (see Algorithm 10). The block sums of level i are updated only once every n^i time step. Therefore, for a total simulation of M time steps, the number of samples is much less for the order- n scheme; for the conventional scheme, we have M samples for all time steps, whereas the order- n scheme has M/n^i samples for the i th block velocity. Naively, one would think that the conventional scheme, therefore, is more accurate. In the conventional scheme, however, the successive samples will have much more correlation and therefore are not independent. To investigate the effect of these correlations on the accuracy of the results, we used the method of Flyvbjerg and Petersen [148] (see Appendix 5.3.3 and Case Study 4). In this method, the standard deviation is calculated as a function of the number of data blocks. If the data are correlated, the standard deviation will increase as a function of the number of blocks until the number of blocks is sufficient that the data in a data block are uncorrelated. If the data are uncorrelated, the standard deviation will be independent of the number of blocks. This limiting value is the standard deviation of interest.

In these simulations, the time step was $\Delta t = 0.001$, and the block length was set to $n = 10$. For both methods the total number of time steps was equal. To calculate the mean-squared displacement, we have used 100,000 samples for all times in the conventional scheme. For the order- n scheme, we have used 100,000 samples for $t \in [0, 0.01]$, 10,000 for $t \in [0.01, 0.1]$, 1,000 for $t \in [0.1, 1]$, etc. This illustrates that the number of samples in the order- n scheme is considerably less than in the conventional scheme. The accuracy of the results, however, turned out to be the same. This is shown in Fig. N.9 for $t = 0.1$ and $t = 1.0$. Since the total number of data blocking operations that can be performed on the data depends on the total number of samples, the number of blocking operations is less for the order- n method. Fig. N.9 shows that for $t = 0.1$ the order- n scheme yields a standard deviation that is effectively constant after three data blocking operations, indicating the samples are independent, whereas the standard deviation using the conventional method shows an increase for the first six to eight data blocking operations. For $t = 1.0$ the order- n method is independent of the number of data blocks, the conventional method only after 10 data blocks. This implies that one has to average over $2^{10} \approx 1000$ successive samples to have two independent data points. In addition, the figure shows that the plateau value of the standard deviation is essentially the same for the two methods, which implies that for this case, the two methods are equally accurate, but as shown below, the order- n method is computationally cheaper.

In Fig. N.10 we compare the CPU requirements of the two algorithms for simulations with a fixed total number of time steps. This figure shows the increase of the total CPU time of the simulation as a function of the total time for which the mean-squared displacement has been calculated. With the order- n scheme, the CPU time should be (nearly) independent of the total time for which we deter-

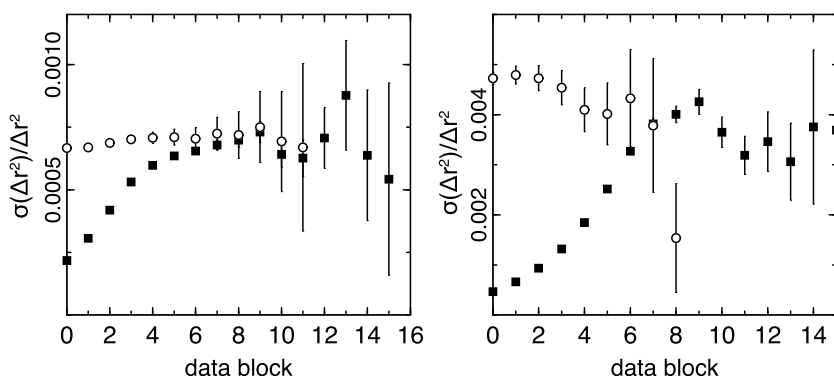


FIGURE N.9 The relative error in the mean-squared displacement as a function of the number of data blocks as defined by Flyvbjerg and Petersen. The figures compare the conventional scheme (solid squares) with the order- n method (open circles) to determine the mean-squared displacement. The right figure is for $t = 0.1$ and the left figure is for $t = 1.0$.

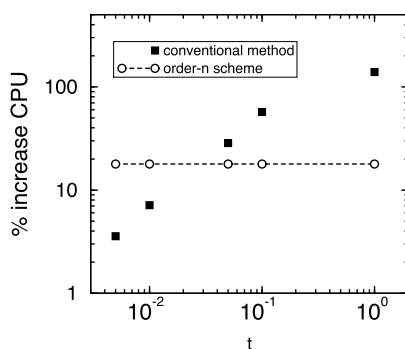


FIGURE N.10 Percentage increase of the total CPU time as a function of the total time for which we determine the mean-squared displacement; comparison of the conventional scheme with the order- n scheme for the same system as is considered in Fig. N.8.

mine the mean-squared displacement, which is indeed what we observe. For the conventional scheme, however, the required CPU time increases significantly for longer times. At $t = 1.0$ the order- n scheme gives an increase of the total CPU time of 17%, whereas the conventional scheme shows an increase of 130%.

This example illustrates that the saving in memory as well as in CPU time of the order- n scheme can be significant, especially if we are interested in the mean-squared displacement at long times.

N.7 Equation of state of the Lennard-Jones fluid

Case Study 7 (Equation of state of the Lennard-Jones fluid). Simulations at constant pressure can also be used to determine the equation of state of a pure component. In such a simulation, the density is determined as a function of the applied pressure and temperature. Fig. N.11 shows that, for the Lennard-Jones fluid, the results of an NPT simulation agree with those obtained in Case Study 1.

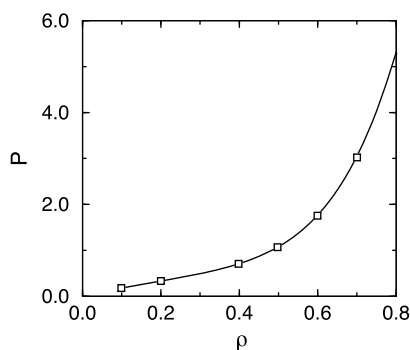


FIGURE N.11 Equation of state of the Lennard-Jones fluid as obtained from NPT simulations; isotherms at $T = 2.0$. The solid line is the equation of state of Johnson et al. [83] and the squares are the results from the simulations ($N = 108$).

In simulations of realistic models of molecular liquids, it is important to verify that the model fluid has the same density as the real fluid under atmospheric pressure. Using NVT simulations, one would require several simulations to determine the density for which the pressure is approximately 1 atm. In an NPT simulation, one would obtain this result in a single simulation. In reduced units, atmospheric pressure is typically much less than one. Hence, a good first estimate of the liquid density at atmospheric pressure follows from an NPT simulation at $P = 0$.³

N.8 Phase equilibria from constant-pressure simulations

Case Study 8 (Phase equilibria from constant-pressure simulations). In Case Studies 1 and 7 NVT or NPT simulations are used to determine the equation of state of a pure substance. In principle, we can construct the liquid-vapor coexistence curve by first fitting the simulation data to an analytical expression, and then determining the liquid and vapor densities for which P , T and μ are the same. Although this route to locate a coexistence curve is quite general, it requires many simulations. More efficient ways to determine the vapor-liquid coexistence curve are discussed in section 6.6. However, in cases where the vapor pressure of the liquid is very small ($\ll 1$ in reduced units), we can determine the density of the coexisting liquid by performing simulations at zero pressure.

To carry out an NPT -simulation at zero pressure it is best to start with a liquid density above the estimated coexistence density. During the simulation, the system will quickly reach the (metastable) density for which $P = 0$. However, starting from a lower density is not recommended as the system might then expand without bounds.

Fig. N.12 shows that the densities obtained in a zero-pressure simulation agree well with the true equation of state of the Lennard-Jones fluid, even up to a reduced temperature of 1, but not as the equilibrium density of the vapor becomes

³ Strictly speaking, the liquid is only metastable at $P = 0$. However, as the nucleation barrier for bubble formation is typically much larger than $k_B T$, the lifetime of this metastable state is usually much longer than the simulation time.

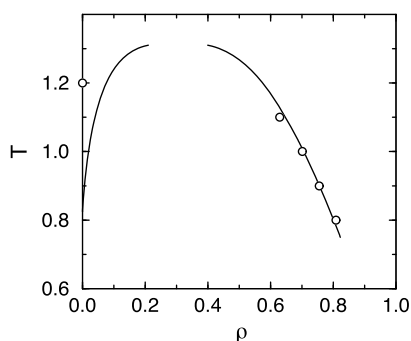


FIGURE N.12 Vapor-liquid coexistence curve of the Lennard-Jones fluid; the solid curve was computed using the Johnson equation of state [83]. The circles indicate the average liquid densities obtained from NPT simulations at $P = 0$. As can be seen from the figure, the zero-pressure method becomes less reliable for $T > 1.2$.

appreciable: for $T > 1$, the zero-pressure simulations predict liquid densities that are too low. Moreover, as the critical temperature T_c is approached, the surface tension tends to zero and, as a consequence, bubble nucleation becomes more likely. Under those conditions, the metastable liquid at $P = 0$ is increasingly likely to evaporate during a simulation. In short: do not use $P = 0$ simulations close to T_c . On a more positive note: not too close to the critical temperature, a reasonable estimate of the liquid density can be obtained by carrying out NPT simulations at $P = 0$.

N.9 Equation of state of the Lennard-Jones fluid - II

Case Study 9 (Equation of state of the Lennard-Jones fluid - II). In Case Studies 1 and 7, we computed the equation of state of a Lennard-Jones fluid, using respectively NVT and NPT simulations. A third way to determine the equation of state is to perform a *Grand-canonical* simulation, imposing the temperature T and the chemical potential μ at constant V , and sample the resulting density and pressure. An example of such a calculation is shown in Fig. N.13.

Grand-canonical simulations are not particularly useful for computing the equation-of-state of a homogeneous fluid, because there will be statistical errors in both the pressure and the density. However, for systems where the pressure itself is ill-defined (e.g., for nano-porous materials), grand-canonical simulations are the method of choice.

N.10 Phase equilibria of the Lennard-Jones fluid

Case Study 10 (Phase equilibria of the Lennard-Jones fluid). Here we use the Gibbs-ensemble technique to determine the vapor-liquid coexistence curve of the Lennard-Jones fluid. In Case Studies 1, 7, and 9, we had already determined parts of the equation of state of this fluid, and in Case Study 8 we had made an estimate of the liquid coexistence-density from a zero-pressure simulation.

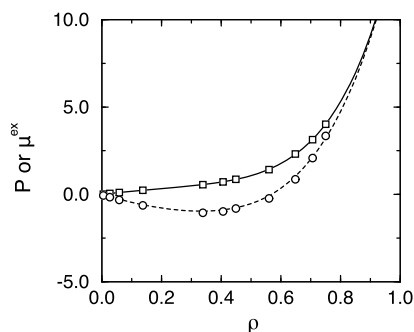


FIGURE N.13 Equation of state of the Lennard-Jones fluid; isotherm at $T = 2.0$. The solid curve represents the equation of state of Johnson et al. [83]; the squares are the results from grand-canonical simulations (with volume $V = 250.047$). The dotted curve is the excess chemical potential as calculated from the equation of state of ref. [83], and the circles are the simulation results. Note that the excess chemical potential is related to the fugacity f through $\beta\mu^{\text{ex}} = \ln(f/\rho)$.

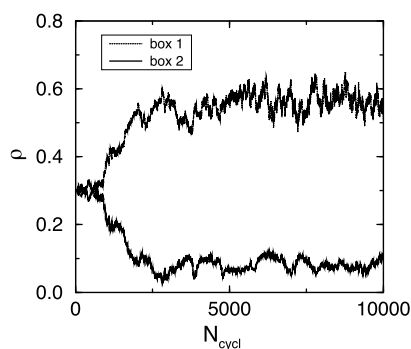


FIGURE N.14 Particle number density in the two boxes of the Gibbs ensemble as a function of the number of Monte Carlo cycles for a system of Lennard-Jones particles; the number of particles was $N = 256$ and temperature $T = 1.2$.

In Fig. N.14, the density of the fluid in the two boxes is plotted as a function of the number of Monte Carlo cycles (as defined in Algorithm 15). The simulation started with equal density in both boxes. During the first 1000 Monte Carlo cycles, the system has not yet “decided” which box would evolve to a liquid density and which to the vapor. After 5000 Monte Carlo cycles, the system seems to have reached equilibrium, and the coexistence properties can be determined.

In Fig. N.15, the phase diagram of the Lennard-Jones as obtained from Gibbs-ensemble simulations is compared with the phase diagram obtained from the equation of state of Johnson et al. [83]. We point out that comparison of the GE phase diagram with the literature data on the critical point (e.g., [212]) is complicated by the fact that different authors use different truncations of the Lennard-Jones potential (see e.g., [80,219]). Such truncations have a large effect on the location of the critical temperature.

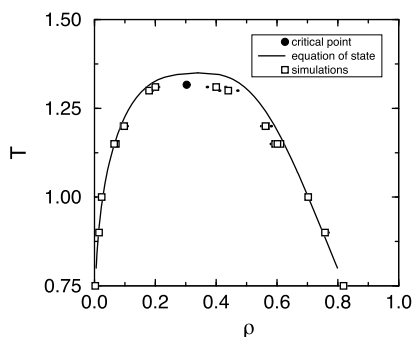


FIGURE N.15 Phase diagram of the Lennard-Jones fluid, using tail correction beyond the cut-off of 2.5σ to mimic the full Lennard-Jones potential, as calculated with the Gibbs-ensemble technique (squares) and equation of state of Johnson et al. (solid curves). The solid circle indicates the estimated critical point.

N.11 Use of Andersen thermostat

Case Study 11 (Use of Andersen thermostat). The Andersen thermostat [180] is arguably the simplest MD thermostat that has been proven to yield a canonical distribution. This implies that both the kinetic energy and the potential energy of the system under consideration are Boltzmann-distributed.

However, the Andersen algorithm conserves neither momentum nor energy, as it attributes new velocities drawn from a Maxwell-Boltzmann distribution to randomly selected particles. We refer to these updates as stochastic “collisions”. The effect of the stochastic collisions is that the dynamical properties of a system with an Andersen thermostat may differ substantially, or even catastrophically, from those of a system without a thermostat, or with a “global” thermostat [256,257].

In the case of diffusion, it is easy to understand intuitively that the Andersen thermostat will decrease the self-diffusion coefficient D^s : a Green-Kubo relation relates D to the integral of the velocity-auto-correlation function (VACF). The longer velocities persist, the larger D . Conversely, any effect that destroys the persistence in the velocity, will decrease D . And destroying the persistence in $\mathbf{v}$ is precisely what the Andersen thermostat does: the higher ν , the frequency of stochastic collisions, the lower D . This effect is illustrated in Fig. N.16.

In practical cases, ν is usually chosen such that the timescale τ_E for the decay of energy fluctuations in a simulation of a system with linear dimension L is comparable to that of thermal fluctuations with a wavelength equal to L , in an unbounded medium: $\tau_E \approx [L^2(N/V)C_P]/\lambda$, where C_P is the heat capacity per molecule (at constant pressure) and λ is the thermal conductivity.

For reasonably large systems, such values of τ_E can only be achieved with rather low collision rates per particle, in which case the effect of collisions on the dynamics may be small [180], but at the same time, the thermostat becomes rather inefficient.

One should not use the Andersen thermostat when computing quantities such as the thermal conductivity or the viscosity.

The reason is that these quantities provide a measure for the rate at which energy or momentum diffuse. But such a description only makes sense if energy and

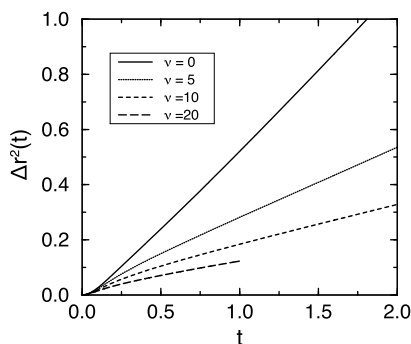


FIGURE N.16 Mean-squared displacement as a function of time for various values of the collision frequency ν of Andersen thermostat. The system studies are a Lennard-Jones fluid ($T = 2.0$ and $N = 108$).

momentum are conserved. With an Andersen thermostat, a local perturbation in the energy or momentum does not diffuse away, and rather, it is screened exponentially. This effect is not described by the (normal) diffusion equation.

Summarizing: do not use the Andersen method when computing transport properties. For static properties, it is fine—and very easy to implement.

Fig. N.16 shows that the frequency of stochastic collisions has a strong effect on the time dependence of the mean-squared displacement. The mean-squared displacement becomes only independent of ν in the limit of very low stochastic collision rates. Yet, all static properties, such as the pressure or potential energy are rigorously independent of the stochastic collision frequency.

N.12 Use of Nosé-Hoover thermostat

Case Study 12 (Use of Nosé-Hoover thermostat). As in Case Study 11, we start by showing that the Nosé-Hoover method reproduces the behavior of a system at constant NVT . In Fig. N.17 we compare the velocity distribution generated by the Nosé-Hoover thermostat with the correct Maxwell-Boltzmann distribution for the same temperature (7.1.1). The figure illustrates that the velocity distribution indeed is independent of the value chosen for the coupling constant Q .

It is instructive to see how the system reacts to a sudden increase in the imposed temperature. Fig. N.18 shows the evolution of the kinetic temperature of the system. After 12,000 time steps, the imposed temperature is suddenly increased from $T = 1$ to $T = 1.5$. The figure illustrates the role of the coupling constant Q . A small value of Q corresponds to a low inertia of the heat bath and leads to rapid temperature fluctuations. A large value of Q leads to a slow, ringing response to the temperature jump.

Next, we consider the effect of the Nosé-Hoover coupling constant Q on the diffusion coefficient. As can be seen in Fig. N.19, the effect is much smaller than in Andersen's method. However, it would be wrong to conclude that the diffusion coefficient is independent of Q . The Nosé-Hoover method simply provides a way to keep the temperature constant more gently than Andersen's method, where

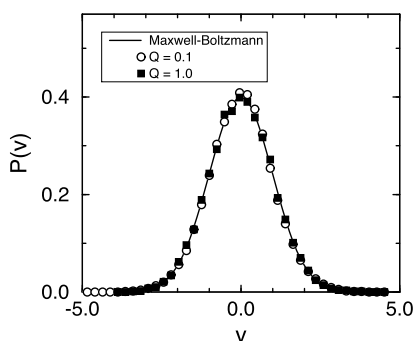


FIGURE N.17 Velocity distribution in a Lennard-Jones fluid ($T = 1.0$, $\rho = 0.75$, and $N = 256$). The solid line is the Maxwell-Boltzmann distribution (7.1.1) the symbols were obtained in a simulation using the Nosé-Hoover thermostat [256].

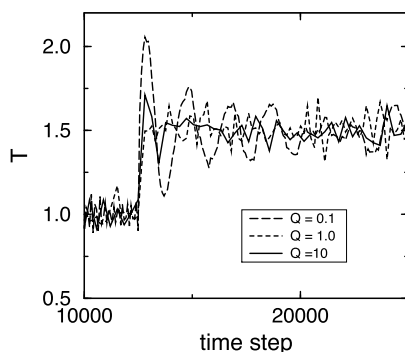


FIGURE N.18 Response of the system to a sudden increase in the imposed temperature. The various lines show the actual temperature of the system (a Lennard-Jones fluid $\rho = 0.75$, and $N = 256$) as a function of the number of time steps for various values of the Nosé-Hoover coupling constant Q .

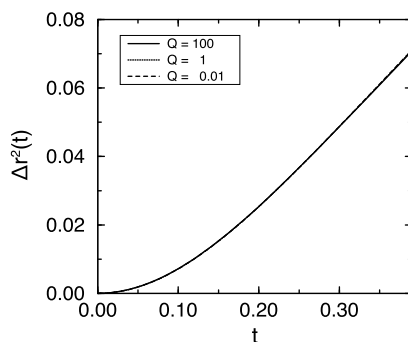


FIGURE N.19 Effect of the coupling constant Q on the mean-squared displacement for a Lennard-Jones fluid ($T = 1.0$, $\rho = 0.75$, and $N = 256$).

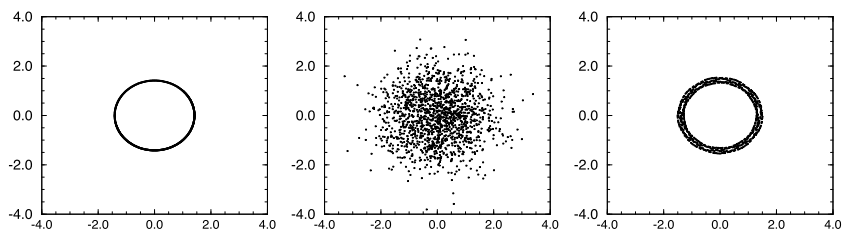


FIGURE N.20 Trajectories of the harmonic oscillator: (from left to right) in the microcanonical ensemble, using the Andersen method and using the Nosé-Hoover method. The y axis is the velocity and the x axis is the position.

particles suddenly get new, random velocities. For the calculations of transport properties, we prefer simple N , V , E simulations.

N.13 Harmonic oscillator (I)

Case Study 13 (Harmonic oscillator (I)). As the equations of motion of the harmonic oscillator can be solved analytically, this model system is often used to test algorithms. However, the harmonic oscillator is also a rather atypical dynamical system, as will become clear when we apply the NH algorithm to this simple model system.

The potential energy function of the harmonic oscillator is

$$u(r) = \frac{1}{2}r^2.$$

The Newtonian equations of motion are

$$\begin{aligned}\dot{r} &= v \\ \dot{v} &= -r.\end{aligned}$$

If we solve the equations of motion of the harmonic oscillator for a given set of initial conditions, we can trace the trajectory of the system in phase space. Fig. N.20 shows a typical phase space trajectory of the harmonic oscillator, in a closed loop, which is characteristic of periodic motion. It is straightforward to simulate a harmonic oscillator at a constant temperature using the Andersen thermostat (see section 7.1.1.1). A trajectory is shown in Fig. N.20. In this case, the trajectories are points that are not connected by lines. This is due to the stochastic collisions with the bath. In this example, we allowed the oscillator to interact with the heat bath at each time step. As a result, the phase space density is a collection of discrete points. The resulting velocity distribution is Gaussian by construction; also, for the positions, we find a Gaussian distribution.

We can also perform a constant-temperature Nosé-Hoover simulation using the algorithm described in SI L.6.2. A typical trajectory of the harmonic oscillator generated with the Nosé-Hoover scheme is shown in Fig. N.20. The most striking feature of Fig. N.20 is that, unlike the Andersen scheme, the Nosé-Hoover method does not yield a canonical distribution in phase space. Even for very long simulations, the entire trajectory would lie in the same ribbon shown in Fig. N.20. Moreover, this band of trajectories depends on the initial configuration.

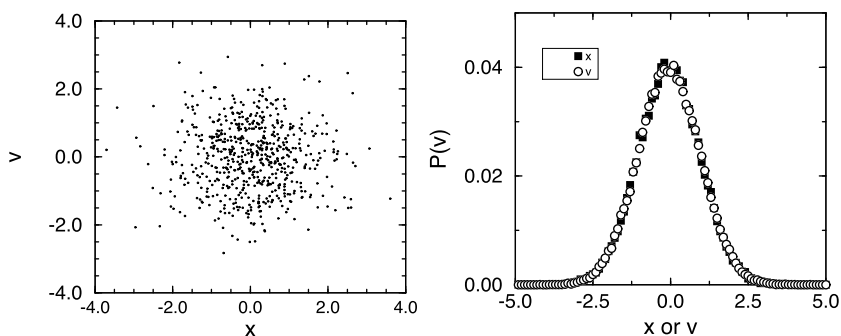


FIGURE N.21 Test of the phase space trajectory of a harmonic oscillator, coupled to a Nosé-Hoover chain thermostat. The left-hand side of the figure shows part of a trajectory: the dots correspond to consecutive points separated by 10,000 time steps. The right-hand side shows the distributions of velocity and position. Due to our choice of units, both distributions should be Gaussians of equal width.

N.14 Nosé-Hoover chain for harmonic oscillator

Case Study 14 (Nosé-Hoover chain for harmonic oscillator). The harmonic oscillator is the obvious model system on which we test the Nosé-Hoover chain thermostat. If we use a chain of two coupling parameters, the equations of motion are

$$\begin{aligned}\dot{r} &= v \\ \dot{v} &= -r - \xi_1 v \\ \dot{\xi}_1 &= \frac{v^2 - T}{Q_1} - \xi_1 \xi_2 \\ \dot{\xi}_2 &= \frac{Q_1 \xi_1^2 - T}{Q_2}.\end{aligned}$$

A typical trajectory generated with the Nosé-Hoover chains is shown in Fig. N.21. The distributions of the velocity and position of the oscillator are also shown in Fig. N.21. Comparison with the results obtained using the Andersen thermostat (see Case Study 13) shows that the Nosé-Hoover chains do generate a canonical distribution, even for the harmonic oscillator.

N.15 Chemical potential: particle-insertion method

Case Study 15 (Chemical potential: particle-insertion method). In this Case Study, we use the Widom test-particle method to determine the excess chemical potential of a Lennard-Jones fluid. The algorithm that we use is a combination of the basic algorithm for performing Monte Carlo simulations at constant N , V , and T (Algorithms 1 and 2) and determining the excess chemical potential (Algorithm 18).

We stress that the tail correction for the chemical potential is similar, but not identical, to that for the potential energy. In the Widom test-particle method, we

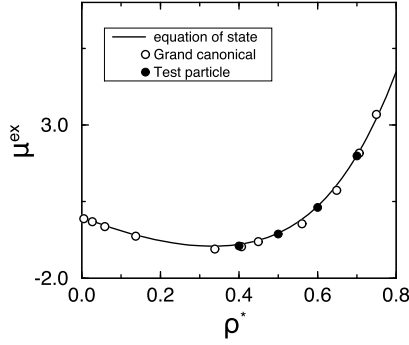


FIGURE N.22 The excess chemical potential of the Lennard-Jones fluid ($T = 2.0$) as calculated from the equation of state, grand-canonical Monte Carlo, and the test particle insertion method.

determine the energy difference:

$$\Delta\mathcal{U} = \mathcal{U}(\mathbf{s}^{N+1}) - \mathcal{U}(\mathbf{s}^N).$$

The tail correction is

$$\begin{aligned} \beta\mu^{\text{tail}} &= \mathcal{U}(\mathbf{s}^{N+1})^{\text{tail}} - \mathcal{U}(\mathbf{s}^N)^{\text{tail}} \\ &= (N+1)u^{\text{tail}}((N+1)/V) - Nu^{\text{tail}}(N/V) \\ &= \left((N+1)\frac{N+1}{V} - N\frac{N}{V} \right) \frac{1}{2} 4\pi \int_{r_c}^{\infty} dr r^2 u(r) \\ &\approx \frac{2N}{V} \frac{1}{2} 4\pi \int_{r_c}^{\infty} dr r^2 u(r) \\ &= 2u^{\text{tail}}(\rho). \end{aligned} \tag{N.15.1}$$

In Case Study 9, we performed a grand-canonical Monte Carlo simulation to determine the equation of state of the Lennard-Jones fluid. In the grand-canonical ensemble the volume, chemical potential, and temperature are imposed; the density is determined during the simulation. Of course, we can also calculate the chemical potential during the simulation, using the Widom method. Fig. N.22 shows a comparison of the imposed and measured chemical potentials.

N.16 Chemical potential: overlapping distributions

Case Study 16 (Chemical potential: overlapping distributions). In Case Study 15, we used the Widom test-particle method to determine the chemical potential of the Lennard-Jones fluid. This method fails at high densities, where it becomes extremely unlikely to insert a particle at a position where $\exp(-\beta\Delta\mathcal{U})$ in Eq. (8.5.5) is non-negligible. Yet it is those unlikely insertions that dominate the average $\langle \exp(-\beta\mathcal{U}) \rangle$. Because favorable insertions are so rare, the number of such events is subject to relatively large statistical fluctuations, and hence our estimate for $\langle \exp(-\beta\mathcal{U}) \rangle$ is noisy. The *overlapping-distribution method* does not remove this

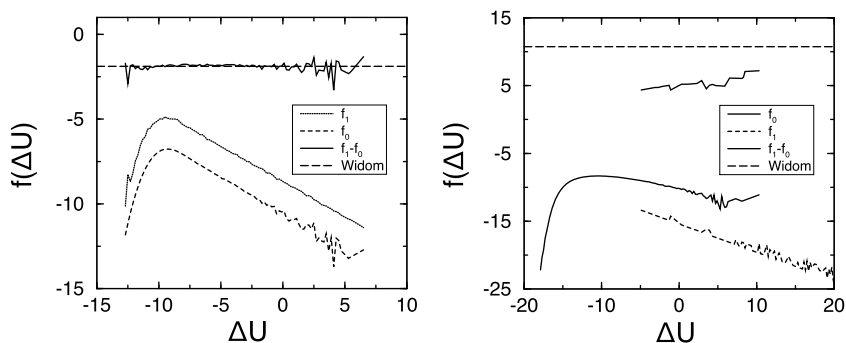


FIGURE N.23 Comparison of the overlapping distribution function method and the Widom particle insertion scheme for measuring the chemical potential of the Lennard-Jones fluid at $T = 1.2$. The solid curve is the particle-insertion result. The dashed curve was obtained using the overlapping distribution method ($\beta\mu_{\text{ex}} = f_1 - f_0$). The units for $\beta\mu_{\text{ex}}$ are the same as for $f(\Delta\mathcal{U})$. The figure on the left corresponds to a moderately dense liquid ($\rho = 0.7$). In this case, the distributions overlap and the two methods yield identical results. The right-hand figure corresponds to a dense liquid ($\rho = 1.00$). In this case, the insertion probability is very low. The distributions f_0 and f_1 hardly overlap, and the two different estimates of $\beta\mu_{\text{ex}}$ do not coincide.

problem, but it provides a good diagnostic tool for detecting such sampling problems.

To implement the overlapping-distribution method, we have to perform two simulations: one simulation using a system of $N + 1$ particles (system 1) and a second system with N particles and one ideal gas particle (system 0). For each of these systems, we determine the distribution of energy differences, Eqs. (8.6.3) and (8.6.4). For system 1, this energy difference $\Delta\mathcal{U}$ is defined as the change in the total energy of the system that would result if one particle, chosen at random, would be transformed into an ideal gas particle. We now make a histogram of the observed values of $\Delta\mathcal{U}$ in this system. This calculation can easily be appended to a standard MC move (Algorithm 2), because in a trial move, we randomly select a particle and compute its interaction energy before a trial move. But that interaction energy is precisely the $\Delta\mathcal{U}$ that we wish to compute. We thus obtain a probability distribution of $\Delta\mathcal{U}$: $p_1(\Delta\mathcal{U})$.

For system 0, we have to determine the energy difference $\Delta\mathcal{U}$, which is the difference in total energy when the ideal gas particle (which could be anywhere in the system) would be turned into an interacting particle. This energy difference equals the energy of a test particle in the Widom method (section 8.5.1). When we determine $p_0(\Delta\mathcal{U})$, at the same time, we can obtain an estimate of the excess chemical potential from the Widom particle insertion method. As explained in the text, it is convenient not to use $p_0(\Delta\mathcal{U})$ and $p_1(\Delta\mathcal{U})$, but the closely related functions $f_0(\Delta\mathcal{U})$ and $f_1(\Delta\mathcal{U})$, defined in Eqs. (8.6.6) and (8.6.7).

In Fig. N.23 we show how $\mu_{\text{ex}}(\Delta\mathcal{U})$ can be obtained (using Eq. (8.6.8)) from a plot of $f_0(\Delta\mathcal{U})$ and $f_1(\Delta\mathcal{U})$, as a function of $\Delta\mathcal{U}$. The results shown in the left part of Fig. N.23 apply to a Lennard-Jones fluid at $\rho = 0.7$, the results on the right are for $\rho = 1.00$.

For the sake of comparison, we have also plotted the results obtained using the Widom particle insertion method. The figure shows that at $\rho = 0.7$ there is a sufficiently large range of energy differences for which the two functions overlap ($-10 < \Delta\mathcal{U} < -5$), in the sense that the noise in both functions is relatively small in this energy range. The result of the overlapping distribution function, therefore, is in good agreement with the results from the Widom method. However, at $\rho = 1.00$, the range of overlap is limited to the wings of the histograms p_0 and p_1 , where the statistical accuracy is poor. As a consequence, our estimate for $\mu_{\text{ex}}(\Delta\mathcal{U})$ is not constant (as it should) but appears to depend on $\Delta\mathcal{U}$. Moreover, the results from the overlapping distribution method are not consistent with the result of the Widom particle insertion method.

Note that two separate simulations are needed to determine the excess chemical potential from the overlapping distribution method. One might think that particle addition and particle removal histograms could be measured in a single simulation of an N -particle system. Such an approach would indeed be correct if there were no difference between the histograms for particle removal from N and $N + 1$ particle systems, but for dense systems containing only a few hundred particles, the system-size dependence of μ_{ex} can be appreciable.

Of course, sometimes simulations on small systems are performed as a preliminary step to larger-scale simulations. Then it is advisable to compute both $p_0(\Delta\mathcal{U})$ and $p_1(\Delta\mathcal{U})$ for the small system in a single simulation, as this allows us to check whether the overlap between the two distributions is likely to be sufficient to justify a larger-scale simulation.

N.17 Solid-liquid equilibrium of hard spheres

Case Study 17 (Solid-liquid equilibrium of hard spheres). In this case study, we locate the solid-liquid coexistence densities of the hard-sphere model. We determine these densities by equating the chemical potential and the pressure of the two phases.

For the liquid phase, we use the equation of state of Speedy [421], which is based on a Padé approximation to simulation data on both the equation of state and the virial coefficients of hard spheres:

$$z_{\text{liquid}} = \frac{P\beta}{\rho} = 1 + \frac{x + 0.076014x^2 + 0.019480x^3}{1 - 0.548986x + 0.075647x^2}.$$

For the solid phase of the hard-sphere model, Speedy proposed the following equation of state [320]:

$$z_{\text{solid}} = \frac{3}{1 - \rho^*} - 0.5921 \frac{\rho^* - 0.7072}{\rho^* - 0.601}, \quad (\text{N.17.1})$$

where $\rho^* = \sigma^3 \rho / \sqrt{2}$. In Fig. N.24, we compare the predictions of this equation of state for the liquid and solid phases with the results from computer simulations of Alder and Wainwright [422] and Adams [171]. As can be seen, the empirical equations of state reproduce the simulation data quite well. To calculate the chemical potential of the liquid phase, we integrate the equation of state (see (9.1.1)) starting

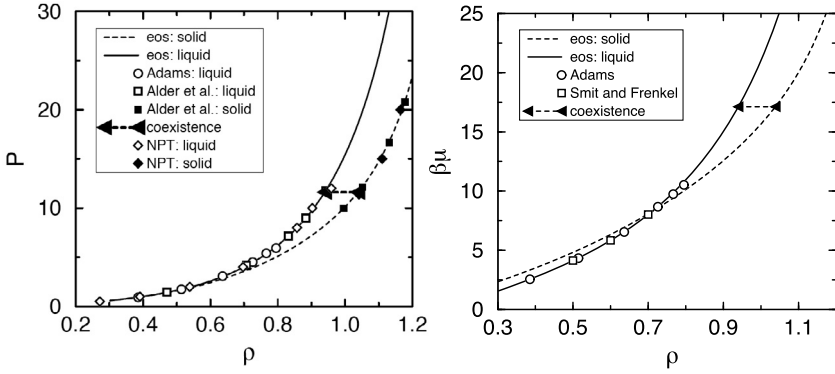


FIGURE N.24 Pressure P (left) and chemical potential μ (right) as a function of the density ρ . The solid curves, showing the pressure and chemical potential of the liquid phase, are obtained from the equation of state of Speedy [421]. The dashed curve gives the pressure of the solid phase as calculated from the equation of state of ref. [320]. The open and filled symbols are the results of computer simulations for the liquid [171,422,423] and solid phases [422], respectively. The coexistence densities are indicated with horizontal lines.

from the dilute gas limit. This yields the Helmholtz free energy as a function of the density. The chemical potential then follows from

$$\beta\mu(\rho) = \frac{\beta G}{N} = \frac{\beta F}{N} + \frac{P}{\rho k_B T}.$$

The free energy per particle of the ideal gas is given by

$$\beta f^{\text{id}}(\rho) = \frac{F^{\text{id}}(\rho)}{N k_B T} = \ln \rho \Lambda^3 - 1,$$

where Λ is the de Broglie thermal wavelength. In what follows, we shall write

$$\beta f^{\text{id}}(\rho) = \ln \rho - 1.$$

That is, we shall work with the usual reduced densities and ignore the additive constant $3 \ln(\Lambda/\sigma)$, as it plays no role in the location of phase equilibria for classical systems.

Fig. N.24 compares the chemical potential that follows from the Hall equation of state with some of the available simulation data (namely, grand-canonical ensemble simulations of [171] and direct calculations of the chemical potential, using the Widom test-particle method [423] (see Chapter 8).

These results show that we have an accurate equation of state for the liquid phase and the solid phase. Since we know the absolute free energy of the ideal gas phase, we can calculate the free energy and hence the chemical potential of the liquid phase. For the solid phase we can use the equation of state to calculate only free energy *differences*; to calculate the absolute free energy we have to determine the free energy at a particular density. To perform this calculation, we use the lattice coupling method.

We must now select the upper limit of the coupling parameter λ ($\lambda_{\max}$) and the values of λ for which we perform the simulation. For sufficiently large values of λ we can calculate $\sum_{i=1}^N (\mathbf{r}_i - \mathbf{r}_{0,i})^2$ analytically, using

$$\langle r^2 \rangle_{\lambda} = \frac{1}{N} \frac{\partial F(\lambda)}{\partial \lambda}.$$

For the noninteracting Einstein crystal, the mean-squared displacement is given by

$$\langle r^2 \rangle_{\lambda} = \frac{3}{2\beta\lambda}. \quad (\text{N.17.2})$$

For a noninteracting Einstein crystal with fixed center of mass, the free energy is given by Eq. (9.2.23), which gives

$$\langle r^2 \rangle_{\text{Ein},\lambda} = \frac{1}{\beta} \frac{3}{2} \frac{N-1}{N} \frac{1}{\lambda}. \quad (\text{N.17.3})$$

In [314] an analytical expression is derived for the case of an interacting Einstein crystal, which reads

$$\begin{aligned} \langle r^2 \rangle_{\lambda} = \langle r^2 \rangle_{\text{Ein},\lambda} - \frac{\beta n}{2} \frac{1}{2a(2\pi\beta\lambda)^{(1/2)} \left(1 - \langle P_{\text{overlap}}^{\text{nn}} \rangle_{\lambda}\right)} \\ \times \left\{ \left[\sigma a - \sigma^2 - 1/(\beta\lambda) \right] \exp \left[-\beta\lambda(a - \sigma)^2/2 \right] \right. \\ \left. + \left[\sigma a + \sigma^2 - 1/(\beta\lambda) \right] \exp \left[-\beta\lambda(a + \sigma)^2/2 \right] \right\}, \end{aligned} \quad (\text{N.17.4})$$

where a is the separation of two nearest neighbors i and j , $a = \mathbf{r}_{0,i} - \mathbf{r}_{0,j}$, σ is the hard-core diameter, and n is the number of nearest neighbors (for example, $n = 12$ for Face Centered Cubic (FCC) and hcp (hexagonal close-packed) solids or $n = 8$ for bcc (body-centered cubic)); $\langle P_{\text{overlap}}^{\text{nn}} \rangle_{\lambda}$ is the probability that two nearest neighbors overlap. Such probability is given by

$$\begin{aligned} \langle P_{\text{overlap}}^{\text{nn}} \rangle_{\lambda} = \frac{\text{erf} \left[(\beta\lambda/2)^{1/2}(\sigma + a) \right] + \text{erf} \left[(\beta\lambda/2)^{1/2}(\sigma - a) \right]}{2} \\ - \frac{\exp[-\beta\lambda(\sigma - a)^2/2] - \exp[-\beta\lambda(\sigma + a)^2/2]}{(2\pi\beta\lambda)^{1/2}a} \end{aligned} \quad (\text{N.17.5})$$

This equation can also be used to correct the free energy of a noninteracting Einstein crystal (9.2.23):

$$\frac{\beta F_{\text{Ein}}(\lambda)}{N} = \frac{\beta F_{\text{Ein}}}{N} + \frac{n}{2} \ln \left(1 - \langle P_{\text{overlap}}^{\text{nn}} \rangle_{\lambda} \right). \quad (\text{N.17.6})$$

We choose $\lambda_{\max}$ such that, for values of λ larger than this maximum value, $\langle r^2 \rangle_{\lambda}$ obeys the analytical expression. Typically, this means that the probability of overlap of two harmonically bound particles should be considerably less than 1%. The results of these simulations are presented in Fig. N.25. This figure shows that if we

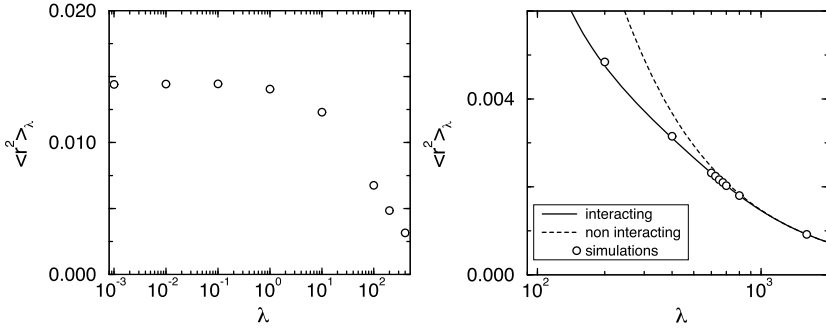


FIGURE N.25 The mean-squared displacement $\langle r^2 \rangle_\lambda$ as a function of the coupling parameter λ for a hard-sphere (FCC) solid of 54 particles (6 layers of 3×3 close-packed atoms at a density $\rho = 1.04$). The figure on the left shows the simulation results for low values of λ , the figure on the right for high values. The solid line takes into account nearest-neighbor interactions (N.17.4); the dashed line assumes a noninteracting Einstein crystal (N.17.3). The open symbols are the simulation results.

rely only on the analytical results of the noninteracting Einstein crystal, we have to take a value for $\lambda_{\max} \approx 1000$ –2000. If we use Eq. (N.17.4) for $\langle r^2 \rangle_\lambda$, $\lambda_{\max} = 500$ –1000 is sufficient.

We should now integrate

$$\frac{\Delta F}{N} = \int_0^{\lambda_{\max}} d\lambda \langle r^2 \rangle_\lambda.$$

In practice, this integration is carried out by numerical quadrature. We therefore must specify the values of λ for which we are going to compute $\langle r^2 \rangle_\lambda$. To improve the accuracy of the numerical quadrature, it is convenient to transform to another integration variable:

$$\frac{\Delta F}{N} = \int_0^{\lambda_{\max}} \frac{d\lambda}{g(\lambda)} g(\lambda) \langle r^2 \rangle_\lambda = \int_{G^{-1}(0)}^{G^{-1}(\lambda_{\max})} d[G^{-1}(\lambda)] g(\lambda) \langle r^2 \rangle_\lambda,$$

where $g(\lambda)$ is an as-yet arbitrary function of λ and $G^{-1}(\lambda)$ is the primitive of the function $1/g(\lambda)$. If we can find a function $g(\lambda)$ such that the integrand, $g(\lambda) \langle r^2 \rangle_\lambda$, is a slowly varying function, we need fewer function evaluations to arrive at an accurate estimate. To do this, we need to have an idea about the behavior of $\langle r^2 \rangle_\lambda$.

For $\lambda \rightarrow 0$, $\langle r^2 \rangle_\lambda \rightarrow \langle r^2 \rangle_0$, which is the mean-squared displacement of an atom around its lattice site in the normal hard-sphere crystal. At high values of λ , where the system behaves like an Einstein crystal, we have $\langle r^2 \rangle_\lambda \rightarrow 3k_B T/(2\lambda)$. This leads to the following guess for the functional form of $g(\lambda)$:

$$g(\lambda) \approx k_B T / \langle r^2 \rangle_\lambda \approx c + \lambda,$$

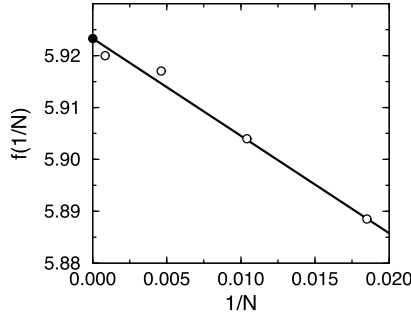


FIGURE N.26 $\beta F^{\text{ex}}/N + \ln(N)/N$ versus $1/N$ for an FCC crystal of hard spheres at a density $\rho\sigma^3 = 1.0409$. The solid line is a linear fit to the data. The coefficient of the $1/N$ term is $-6.0(2)$, and the intercept (i.e., the infinite system limit of $\beta F^{\text{ex}}/N$) is equal to $5.91889(4)$.

where $c = k_B T / \langle r^2 \rangle_0$. Here, $\langle r^2 \rangle_0$ can be estimated from Fig. N.25. The value of c clearly depends on density (and temperature). For $\rho = 1.04$, extrapolation to $\lambda \rightarrow 0$ gives $\langle r^2 \rangle_0 \approx 0.014$, which gives $c = 70$. If we use this function $g(\lambda)$, the free energy difference is calculated from

$$\frac{\Delta F}{N} = \int_{\ln c}^{\ln(\lambda_{\text{max}} + c)} d[\ln(\lambda + c)] (\lambda + c) \langle r^2 \rangle_{\lambda}.$$

For the numerical integration, we use a n -point Gauss-Legendre quadrature [424]. As the integrand is a smooth function, a 10-point quadrature is usually adequate. As discussed in section 9.2.5, the resulting free energy still depends (slightly) on the system size. An example of the system-size dependence of the excess free energy of a hard-sphere crystal is shown in Fig. N.26 [425]. From this figure, we can estimate the excess free energy of the infinite system to be $\beta f^{\text{ex}} = 5.91889(4)$. This is in good agreement with the estimate of Frenkel and Ladd, $\beta f^{\text{ex}} = 5.9222$ [314].

Once we have one value of the absolute free energy of the solid phase at a given density, we can compute the chemical potential of the solid phase at any other density, using the equation of state of Speedy (see Fig. N.24). The coexistence densities follow from the condition that the chemical potentials and pressures in the coexisting phases should be equal.

Using the value of $5.91889(4)$ from [425] for the solid at $\rho = 1.04086$, we arrive at a freezing density $\rho_l = 0.9391$ and a melting density $\rho_s = 1.0376$. At coexistence, the pressure is $P_{\text{coex}} = 11.567$ and the chemical potential is $\mu_{\text{coex}} = 17.071$. In fact, as we shall argue below, the presence of vacancies in the equilibrium crystal lowers the coexistence pressure slightly: $P_{\text{coex}} = 11.564$. These results are in surprisingly good agreement with the original data of Hoover and Ree [307], who obtained an estimate for the solid-liquid coexistence densities $\rho_s = 1.041 \pm 0.004$ and $\rho_l = 0.943 \pm 0.004$ at a pressure 11.70 ± 0.18 .

The free energy difference between the FCC and HCP for large hard-sphere crystals at melting is very close to 0, but the FCC structure appears to be the more stable phase [303,412,426,427].

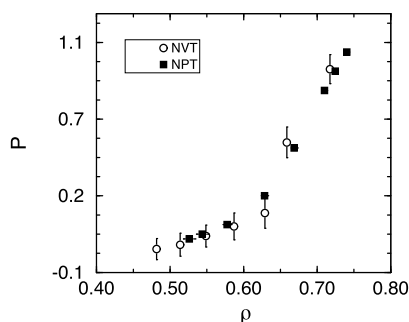


FIGURE N.27 Equation of state of an eight-bead Lennard-Jones chain as obtained from N, V, T and N, P, T simulations using the configurational-bias Monte Carlo scheme. The simulations are performed with 50 chains at a temperature $T = 1.9$.

N.18 Equation of state of Lennard-Jones chains

Case Study 18 (Equation of state of Lennard-Jones chains). To illustrate the configurational-bias Monte Carlo technique described in this section, we determine the equation of state of a system consisting of eight-bead chains of Lennard-Jones particles. The nonbonded interactions are described by a truncated and shifted Lennard-Jones potential. The potential is truncated at $R_c = 2.5\sigma$. The bonded interactions are described with a harmonic spring

$$u^{\text{vib}}(l) = \begin{cases} 0.5k_{\text{vib}}(l - 1)^2 & 0.5 \leq l \leq 1.5 \\ \infty & \text{otherwise} \end{cases},$$

where l is the bond length, the equilibrium bond length has been set to 1, and $k_{\text{vib}} = 400$.

The simulations are performed in cycles. In each cycle, we perform on average N_{dis} attempts to displace a particle, N_{cbmc} attempts to (partly) regrow a chain, and N_{vol} attempts to change the volume (only in the case of N, P, T simulations). If we regrow a chain, the configurational-bias Monte Carlo scheme is used. In this move we select at random the monomer from which we start to regrow. If this happens to be the first monomer, the entire molecule is regrown at a random position. For all the simulations, we used eight trial orientations. The lengths of trial bonds are generated with a probability prescribed by the bond-stretching potential (see Case Study 19).

In Fig. N.27 the equation of state as obtained from N, V, T simulations is compared with one obtained from N, P, T simulations. This isotherm is well above the critical temperature of the corresponding monomeric fluid ($T_c = 1.085$, see Fig. 3.3), but the critical temperature of the chain molecules is appreciably higher [512].

N.19 Generation of trial configurations of ideal chains

Case Study 19 (Generation of trial configurations of ideal chains). In section 12.2.3, we emphasized the importance of efficiently generating trial segments for mole-

cules with strong intramolecular interactions in a **CBMC** simulation. In this Case Study, we quantify this. We consider the following bead-spring model of a polymer. The nonbonded interactions are described with a Lennard-Jones potential, and the bonded interactions with a harmonic spring:

$$u^{\text{vib}}(l) = \begin{cases} 0.5k_{\text{vib}}(l-1)^2 & 0.5 \leq l \leq 1.5 \\ \infty & \text{otherwise} \end{cases},$$

where l is the bond length, the equilibrium bond length has been set to 1, and $k_{\text{vib}} = 400$. The bonded interaction is only the bond stretching. The external (non-bonded) interactions are the Lennard-Jones interactions. We consider the following two schemes of generating a set of trial positions:

1. Generate a random orientation with bond length uniformly distributed in the spherical shell between limits chosen such that they bracket all acceptable bond lengths. For instance, we could consider limits that correspond to a 50% stretching or compression of the bond. In that case, the probability of generating bond length l is given by

$$p_1(l) \begin{cases} \propto C dl \propto l^2 dl & 0.5 \leq l \leq 1.5 \\ 0 & \text{otherwise} \end{cases}.$$

2. Generate a random orientation and the bond length prescribed by the bond-stretching potential (as described in Algorithm 25). The probability of generating bond length l with this scheme is

$$p_2(l) \begin{cases} \propto C \exp[-\beta u^{\text{vib}}(l)] dl = C \exp[-\beta u^{\text{vib}}(l)] l^2 dl & 0.5 \leq l \leq 1.5 \\ 0 & \text{otherwise} \end{cases}.$$

Let us consider a case in which the system consists of ideal chains. Ideal chains are defined (see section 12.2.3) as chains having only *bonded* interactions.

Suppose we use method 1 to generate the set of k trial orientations with bond lengths $l_1, \dots, l_k$, then the Rosenbluth factor, W , for atom i is given by

$$w_i(n) = \sum_{j=1}^k \exp[-\beta u^{\text{vib}}(l_j)].$$

The Rosenbluth factor of the entire chain is

$$W(n) = \prod_{i=1}^{\ell} w_i(n).$$

For the old conformation, a similar procedure is used to calculate its Rosenbluth factor:

$$W(o) = \prod_{i=1}^{\ell} w_i(o).$$

In absence of external interactions, the Rosenbluth factor of the first atom is defined to be $w_1 = k$.

In the second scheme, we generate the set of k trial orientations with a bond length distribution $p_2(l)$. If we use this scheme, we have to consider only the external interaction. Since, for an ideal chain, the external interactions are, by definition, 0, the Rosenbluth factor for each atom is given by

$$w_i^{\text{ext}}(n) = \sum_{j=1}^k \exp[-\beta u^{\text{ext}}(l_j)] = k,$$

and similarly, for the old conformation

$$w_i^{\text{ext}}(o) = k.$$

Hence, the Rosenbluth weight is the same for the new and the old conformations:

$$W^{\text{ext}}(n) = \prod_{i=1}^{\ell} w_i^{\text{ext}}(n) = k^{\ell}$$

and

$$W^{\text{ext}}(o) = \prod_{i=1}^{\ell} w_i^{\text{ext}}(o) = k^{\ell}.$$

The acceptance rule for the first scheme is

$$\text{acc}(o \rightarrow n) = \min[1, W(n)/W(o)]$$

and for the second scheme is

$$\text{acc}(o \rightarrow n) = \min[1, W^{\text{ext}}(n)/W^{\text{ext}}(o)] = 1.$$

Inspection of these acceptance rules shows that, in the second scheme, *all* configurations generated are accepted, whereas in the first scheme, this probability depends on the bond-stretching energy and, therefore, will be less than 1. Hence, it is clearly useful to employ the second scheme.

To show that the results of schemes 1 and 2 are indeed equivalent, we compare the distribution of the bond length of the chain and the distribution of the radius of gyration in Fig. N.28. The figure shows that the results for the two methods are indeed indistinguishable. The efficiency of the two methods, however, is very different. In Table N.1, the difference in acceptance probability is given for some values of the bond-stretching force constant and various chain lengths. The table shows that if we use method 1 and generate a uniformly distributed bond length, we need to use at least 10 trial orientations to have a reasonable acceptance for chains longer than 20 monomers. Note that the corresponding table for the second method has a 100% acceptance for all values of k independent of the chain length.

Most of the simulations, however, do not involve ideal chains but chains with external interactions. For chains with external interactions, the first method performs even worse. First of all, we generate the chains the same way as in the case

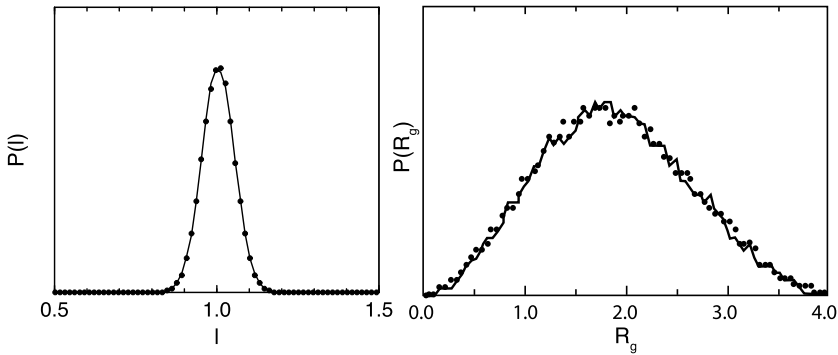


FIGURE N.28 Comparison of methods 1 and 2 for the distribution of bond lengths l (left) and the distribution of the radius of gyration R_g (right). The solid lines represent the results for method 1, and the dots for method 2 ($\ell = 5$ and $k = 5$).

TABLE N.1 Probability of acceptance (%) for ideal chains using uniformly distributed bond lengths (method 1), where ℓ is the chain length and k is the number of trial orientations. The value for the spring constant is $k_{\text{vib}} = 400$ (see [440]). For method 2, the acceptance would have been 100% for all values of k and ℓ .

k	$\ell = 5$	$\ell = 10$	$\ell = 20$	$\ell = 40$	$\ell = 80$	$\ell = 160$
1	0.6	$\ll 0.01$	$\ll 0.01$	$\ll 0.01$	$\ll 0.01$	$\ll 0.01$
5	50	50	10	$\ll 0.01$	$\ll 0.01$	$\ll 0.01$
10	64	58	53	42	$\ll 0.01$	$\ll 0.01$
20	72	66	60	56	44	$\ll 0.01$
40	80	72	67	62	57	40
80	83	78	72	68	62	60

of the ideal chains. The bonded interactions are the same, and we need to generate at least the same number of trial directions to get a reasonable acceptance. In addition, if there are external interactions, we have to calculate the nonbonded interactions for *all* of these trial positions. The calculation of the nonbonded interactions takes most of the CPU time, yet, in the first method, most of the trial orientations are doomed to be rejected solely on the basis of the bonded energy. These two reasons make the second scheme much more attractive than the first.

N.20 Recoil growth simulation of Lennard-Jones chains

Case Study 20 (Recoil growth simulation of Lennard-Jones chains). To illustrate the [Recoil Growth \(RG\)](#) method, we make a comparison between this method and [Configurational-Bias Monte Carlo \(CBMC\)](#). Consider 20 Lennard-Jones chains of length 15. The monomer density is $\rho = 0.3$ at temperature $T = 6.0$. Two bonded monomers have a constant bond length of 1.0, while three successive particles have a constant bond angle of 2.0 radians.

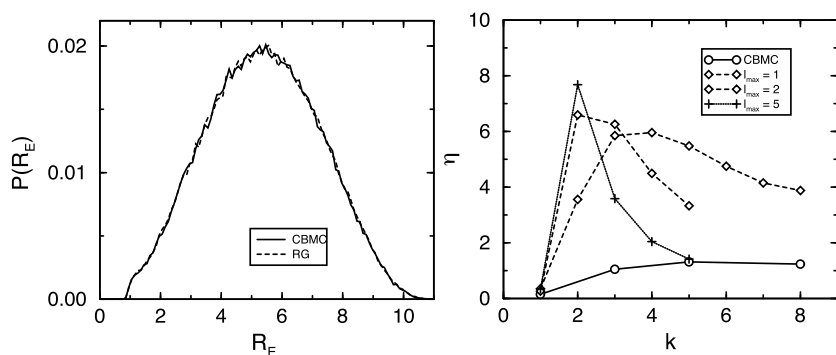


FIGURE N.29 Comparison of configurational-bias Monte Carlo (CBMC) with recoil growth for the simulation of Lennard-Jones chains of length 15. The left figure gives the distribution of the end-to-end distance (R_E). In the right figure the efficiency (η) is a function of the number of trial directions (k) for different recoil lengths ($l_{\max}$) as well as for CBMC.

In Fig. N.29, the distribution of the end-to-end vector, R_E , of the chain is plotted. In this figure, we compare the results from a CBMC and a RG. Since both methods generate a Boltzmann distribution of conformations, the results are identical (as they should be).

For this specific Case Study, we have compared the efficiency, η , of the two methods. The efficiency is defined as the number of accepted trial moves per amount of CPU time. For CBMC we see that the efficiency increases as we increase k , the number of trial orientations, from 1 to 4. From 4 to 8, the efficiency is more or less constant, and above 8, a decrease in the efficiency is observed.

In the RG scheme, we have two parameters to optimize: the number of trial orientations k and the recoil length $l_{\max}$. If we use only one trial orientation, recoiling is impossible since there are no other trial orientations. If we use a recoil length of 1, the optimum number of trial orientations is 4, and for larger recoil lengths, the optimum is reached with fewer trial orientations. Interestingly, the global optimum is 2 trial orientations and a recoil length of 3–5. In this regime, the increase in CPU time associated with a larger recoil length is compensated by a higher acceptance. In the present study, optimal RG was a factor 8 more efficient than optimal CBMC.

Case Study 21 (Parallel tempering of a single particle). As a trivial illustration of the power of parallel tempering, we consider a single particle moving an external potential as shown in Fig. N.30(left):

$$U(x) = \begin{cases} \infty & x < -2 \\ 1 \times (1 + \sin(2\pi x)) & -2 \leq x \leq -1.25 \\ 2 \times (1 + \sin(2\pi x)) & -1.25 \leq x \leq -0.75 \\ 3 \times (1 + \sin(2\pi x)) & -0.75 \leq x \leq 0.75 \\ 4 \times (1 + \sin(2\pi x)) & 0.75 \leq x \leq 1.75 \\ 5 \times (1 + \sin(2\pi x)) & 1.75 \leq x \leq 2 \\ \infty & x > 2 \end{cases} \quad (\text{N.20.1})$$

We place the particle initially in the left-most potential energy well.

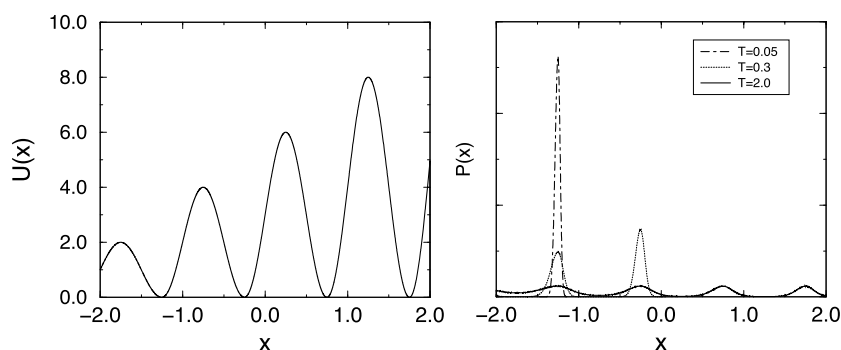


FIGURE N.30 (Left) Potential energy ($U(x)$) as a function of the position x . (Right) Probability ($P(x)$) of finding a particle at position x for various temperatures (T) as obtained from ordinary Monte Carlo simulations. The lower-temperature systems are not (or barely) able to cross the energy barriers separating the wells.

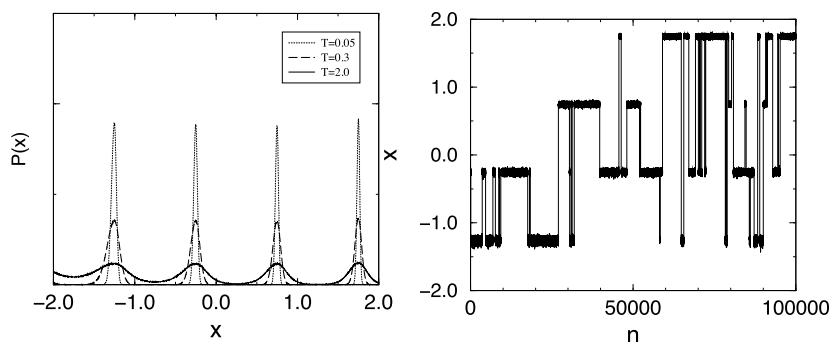


FIGURE N.31 (Left) Probability ($P(x)$) of finding a particle at position x for various temperatures (T) using parallel tempering. (Right) Position (x) as a function of the number of Monte Carlo trial moves (n) for $T = 0.05$. To avoid cluttering this figure with too many jumps, we have only used 0.05% swap moves here.

We first use normal Metropolis MC at three different temperatures ($T = 0.05$, 0.3 , and 2.0). At the lowest temperature ($T = 0.05$), the particle is basically trapped in its initial potential energy well during the entire simulation, whereas for the highest temperature ($T = 2.0$), it can explore all wells.

Next, we apply parallel tempering, that is: we allow for temperature swaps between the three systems (see Fig. N.31). Due to the temperature-swap moves, the systems now equilibrate rapidly at all three temperatures. The difference is particularly strong for the probability distribution at the lowest temperature. In the present parallel-tempering simulation, we consider two types of trial moves:

1. Particle displacement: we randomly select one of the three temperatures and carry out a trial displacement Δ of a randomly selected particle at that temperature, choosing the trial displacement Δ for a random distribution, uniform between -0.1 and 0.1 . The acceptance of this trial displacement is determined

by the conventional Metropolis MC rule

$$\text{acc}(o \rightarrow n) = \min\{1, \exp[-\beta(U(n) - U(o))]\}. \quad (\text{N.20.2})$$

2. Temperature swapping. The trial move consists of attempting to swap two randomly selected neighboring temperatures (T_i and T_j). Such a trial move is accepted with a probability given by Eq. (13.1.2)

$$\text{acc}(o \rightarrow n) = \min\left(1, \exp\left[(\beta_j - \beta_i) \times (U_j - U_i)\right]\right). \quad (\text{N.20.3})$$

We are free to choose the relative rate of displacement and swap moves. In the present example, we used 10% swap moves and 90% particle displacements, as suggested in ref. [550]. Note, however, that other choices are possible.⁴ As can be seen in Fig. N.31, parallel tempering results in a dramatic improvement in the sampling of configuration space at the lowest temperature.

N.21 Multiple time step versus constraints

Case Study 22 (Multiple time step versus constraints). In this Case Study, we consider a system of diatomic Lennard-Jones molecules. We compare two models: the first model uses a fixed bond length l_0 between the two atoms of a molecule. In the second model, we use a bond-stretching potential given by

$$U_{\text{bond}}(l) = \frac{1}{2}k_b(l - l_0)^2,$$

where l is the distance between the two atoms in a molecule. In the simulations, we used $k_b = 50000$ and $l_0 = 1$. In addition to the bond-stretching potential, all non-bonded atoms interact via a Lennard-Jones potential. The total number of diatomics was 125, and the box length 7.0 (in the usual reduced units). The Lennard-Jones potential was truncated at $r_c = 3.0$, while $T = 3.0$. The equations of motion are solved using bond constraints for the first model, while multiple time steps were used for the second model. All simulations were performed in the NVE ensemble.

It is interesting to compare the maximum time steps that can be used to solve the equations of motion for these two methods. As a measure of the accuracy with which the equations of motion are solved, we compute the average deviation of the initial energy, which is defined by Martyna et al. [623] as

$$E = \frac{1}{N_{\text{step}}} \sum_{i=1}^{N_{\text{step}}} \left| \frac{E(i \Delta t) - E(0)}{E(0)} \right|,$$

in which $E(i)$ is the total energy at time i .

⁴ There are even algorithms that use an infinite swap rate [551], including all possible permutations of the temperatures. However, such algorithms (see also [552]) do not scale well with the number of distinct temperatures.

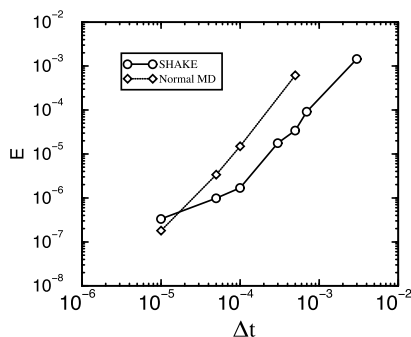


FIGURE N.32 Comparison of the energy fluctuations as a function of the time step for a normal MD simulation with a harmonic bond potential and a constrained MD simulation with the SHAKE algorithm.

For the bond constraints, we use the SHAKE algorithm [606] (see also section 14.1). In the SHAKE algorithm, the bond lengths are exactly fixed at l_0 using an iterative scheme. In Fig. N.32 (left), the energy fluctuations are shown as a function of the time step. Normally one tolerates a noise level in E of $\mathcal{O}(10^{-5})$, which would correspond to a time step of 2×10^{-4} for the first model. This should be compared with a single-time-step Molecular Dynamics simulation using the second model. A similar energy noise level can be obtained with a time step of 9×10^{-5} , which is a factor 2 smaller.

To apply the multiple-time-step algorithm, we have to separate the intermolecular force into a short-range and a long-range part. In the short-range part, we include the bond-stretching potential and the short-range part of the Lennard-Jones potential. To make a split in the Lennard-Jones potential, we use a simple switching function $S(r)$:

$$U_{LJ}(r) = U_{\text{short}}(r) + U_{\text{long}}(r)$$

$$U^{\text{short}}(r) = S(r) \times U_{LJ}(r)$$

$$U^{\text{long}}(r) = [1 - S(r)] U_{LJ}(r),$$

where

$$S(r) = \begin{cases} 1 & 0 < r < r_c - \lambda \\ 1 + \gamma^2 (2\gamma - 3) & r_m - \lambda < r < r_c \\ 0 & r_m < r < r_c \end{cases}$$

and

$$\gamma = \frac{r - r_m + \lambda}{\lambda}. \quad (\text{N.21.1})$$

In fact, there are other ways to split the total potential function [624,625]. We have chosen $\lambda = 0.3$ and $r_m = 1.7$. To save CPU time, a list is made of all the atoms that are close to each other (see Appendix I for details); therefore, the calculation of the

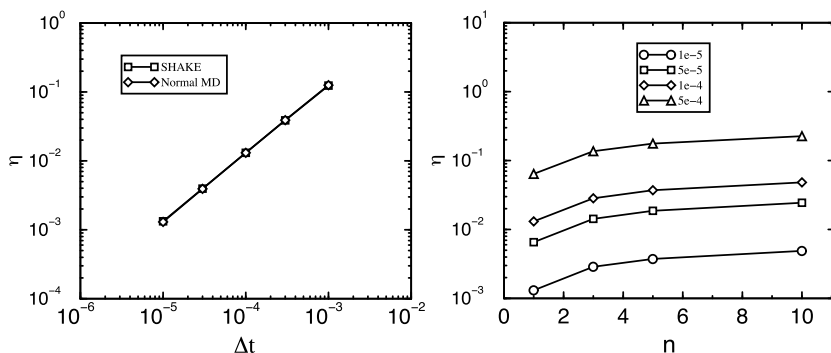


FIGURE N.33 Comparison of the efficiency η for bond constraints (SHAKE) with normal molecular dynamics (left) and multiple time steps (right). The left figure gives the efficiency as a function of the time step and the right figure as a function of the number of small time steps n , $\Delta t = n\delta t$, where the value of δt is given in the symbol legend.

short-range forces can be done very efficiently. For a noise level of 10^{-5} , one is able to use $\delta t = 10^{-4}$ and $n = 10$, giving $\Delta t = 10^{-3}$.

To compare the different algorithms in a consistent way, we compare in Fig. N.33 the efficiency of the various techniques. The efficiency η is defined as the length of the simulation (time step times the number of integration steps) divided by the amount of CPU time that was used. In the figure, we have plotted η for all simulations from Fig. N.32. For an energy noise level of 10^{-5} , the SHAKE algorithm is twice as efficient than normal MD ($n = 1$). This means that hardly any CPU time is spent in the SHAKE routine. However, the MTS algorithm is still two times faster ($n = 10$, $\delta t = 10^{-4}$) at the same efficiency.

N.22 Ideal gas particle over a barrier

Case Study 23 (Ideal gas particle over a barrier). To illustrate the “Bennett-Chandler” approach for calculating crossing rates, we consider an ideal gas particle moving in an external field. This particle is constrained to move on the dimensional potential surface shown in Fig. N.34. This example is rather unphysical because the moving particle cannot dissipate its energy. As a consequence, the motion of the particle is purely ballistic. We assume that, far away on either side of the barrier, the particle can exchange energy with a thermal reservoir. Transition state theory predicts a crossing rate given by Eq. (15.2.2):

$$k_{A \rightarrow B}^{\text{TST}} = \frac{1}{2} |\dot{q}| \frac{\exp[-\beta u(q^*)]}{\int_{-\infty}^{q^*} dq \exp[-\beta u(q)]} = \sqrt{\frac{k_B T}{2\pi m}} \frac{\exp[-\beta u(q^*)]}{\int_{-\infty}^{q^*} dq \exp[-\beta u(q)]}. \quad (\text{a})$$

If we choose the dividing surface q_1 (see Fig. N.34) at the top of the barrier ($q_1 = q^*$) none of the particles that start off with a positive velocity will return to the reactant state. Hence, there is no recrossing of the barrier and transition state theory is exact for this system.

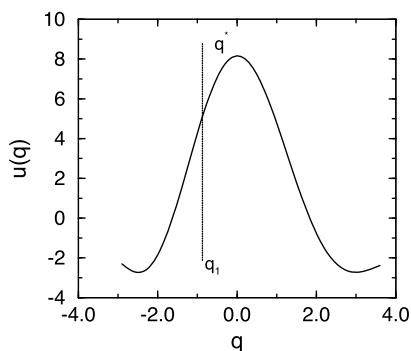


FIGURE N.34 Potential-energy barrier for an ideal gas particle; if the particle has a position to the left of the dividing surface q_1 the particle is in state A (reactant). The region to the right of the barrier is designated as product B . The top of the barrier is denoted by q^* ($q^* = 0$).

Note that transition state theory (Eq. (a)) predicts a rate constant that depends on the location of the dividing surface. In contrast, the Bennett-Chandler expression for the crossing rate is independent of location of the dividing surface (as it should be). To see this, consider the situation that the dividing surface is chosen to be the left of the top of the barrier (i.e., at $q_1 < q^*$). The calculation of the crossing rate according to Eq. (15.2.3) proceeds in two steps. First we calculate the relative probability of finding a particle at the dividing surface. And then we need to compute the probability that a particle that starts with an initial velocity $\dot{q}$ from this dividing surface will, in fact, cross the barrier. The advantage of the present example is that this probability can be computed explicitly. According to Eq. (15.2.4), the relative probability of finding a particle at q_1 is given by

$$\frac{\langle \delta(q - q_1) \rangle}{\langle \theta(q_1 - q) \rangle} = \frac{\exp[-\beta u(q_1)]}{\int_{-\infty}^{q_1} dq \exp[-\beta u(q)]}. \quad (\text{b})$$

If the dividing surface is not at the top of the barrier, then the probability of finding a particle will be higher at q_1 than at q^* , but the fraction of the number of particles that actually cross the barrier will be less than predicted by transition state theory. It is convenient to introduce the time-dependent transmission coefficient, κ . $\kappa = \kappa(t)$, defined as the ratio

$$\kappa(t) \equiv \frac{k_{A \rightarrow B}(t)}{k_{A \rightarrow B}^{\text{TST}}} = \frac{\langle \dot{q}(0) \delta(q(0) - q_1) \theta(q(t) - q_1) \rangle}{0.5 \langle |\dot{q}(0)| \rangle}. \quad (\text{c})$$

The behavior of $\kappa(t)$ is shown in Fig. N.35 for various choices of q_1 . The figure shows that for $t \rightarrow 0$ $\kappa(t) = 1$, and that for different values of q_1 we get different plateau values. The reason $\kappa(t)$ decays from its initial value is that particles that start off with too little kinetic energy cannot cross the barrier and recross the dividing surface (q_1). The plateau value of $\kappa(t)$ provides us with the correction that has to be applied to the crossing rate predicted by transition state theory. Hence, we see that as we change q_1 , the probability of finding a particle at q_1 goes up, and the transmission coefficient goes down. But, as can be seen from Fig. N.35, the actual

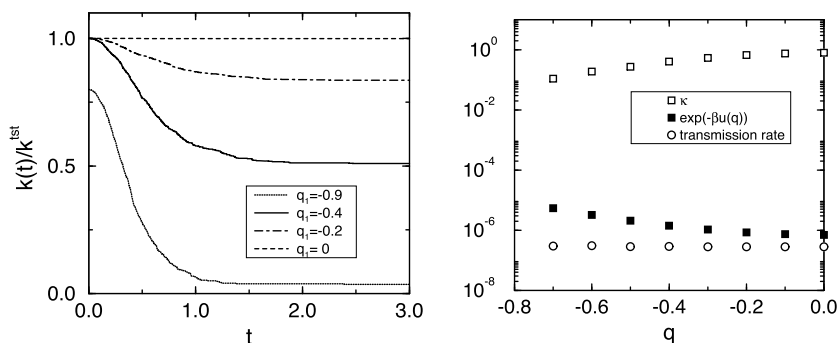


FIGURE N.35 Barrier recrossing: the left figure gives the transmission coefficient as a function of time for different values of q_1 . The right-hand figure shows, in a single plot, the probability density of finding the system at $q = q_1$ (solid squares), the transmission coefficient κ (open squares), and the overall crossing rate (open circles), all plotted as a function of the location of the dividing surface. Note that the overall crossing rate is independent of the choice of the dividing surface.

crossing rate (which is proportional to the product of these two terms) is independent of q_1 , as it should be. Now consider the case that $q_1 > q^*$. In that case, all particles starting with positive $\dot{q}$ will continue to the product side. But now there is also a fraction of the particles with negative $\dot{q}$ that will proceed to the product side. These events will give a negative contribution to κ . And the net result is that the transmission coefficient will again be less than predicted by transition state theory. Hence, the important thing is not if a trajectory ends up on the product side, but if it starts on the reactant side and proceeds to the product side. In a simulation, it is therefore convenient always to compute trajectories in pairs: for every trajectory starting from a given initial configuration with a velocity $\dot{q}$, we also compute the time-reversed trajectory, i.e., the one starting from the same configuration with a velocity $-\dot{q}$. If both trajectories end up on the same side of the barrier then their total contribution to the transmission coefficient is clearly zero. Only if the forward and time-reversed trajectories end up on different sides of the barrier, do we get a contribution to κ . In the present (ballistic) case, this contribution is always positive. But in general, this contribution can also be negative (namely, if the initial velocity at the top of the barrier is not in the direction where the particle ends up).

We chose this simple ballistic barrier-crossing problem because we can easily show explicitly that the transmission rate is independent⁵ of the location of q_1 . We start with the observation that the sum of the kinetic and potential energies of a particle that crosses the dividing surface q_1 is constant. Only those particles that have sufficient kinetic energy can cross the barrier. We can easily compute the long-time limit of $\langle \dot{q}(0)\theta(q(t) - q_1) \rangle$:

$$\langle \dot{q}(0)\theta(q(\infty) - q_1) \rangle = \sqrt{\frac{m\beta}{2\pi}} \int_{v_c}^{\infty} dv v \exp(-\beta m v^2/2)$$

⁵ The general proof that the long-time limit of the crossing rate is independent of the location of the dividing surface was given by Miller [630].

$$= \sqrt{\frac{1}{2\pi m\beta}} \exp\left(-\frac{1}{2}\beta m v_\epsilon^2\right),$$

where v_ϵ is the minimum velocity needed to achieve a successful crossing. v_ϵ is given by

$$\frac{1}{2}m v_\epsilon^2 + u(q_1) = u(q^*).$$

It then follows that

$$\langle \dot{q}(0) \theta(q(\infty) - q_1) \rangle = \sqrt{\frac{1}{2\pi m\beta}} \exp\{-\beta[u(q^*) - u(q_1)]\}.$$

This term exactly compensates the Boltzmann factor, $\exp(-\beta u(q_1))$, associated with the probability of finding a particle at q_1 . Hence, we have shown that the overall crossing rate is given by Eq. (a), independent of the choice of q_1 .

The reader may wonder why it is important to have an expression for the rate constant that is independent of the precise location of the dividing surface. The reason is that, although it is straightforward to find the top of the barrier in a one-dimensional system, the precise location of the saddle point in a reaction pathway of a many-dimensional system is usually difficult to determine. With the Bennett-Chandler approach it is not necessary to know the exact location of the saddle point. Still, it is worth trying to get a reasonable estimate, as the statistical accuracy of the results is best if the dividing surface is chosen close to the true saddle point.

The nice feature of the Bennett-Chandler expression for barrier-crossing rates is that it allows us to compute rate constants under conditions where barrier recrossings are important, for instance, if the motion over the top of the barrier is more diffusive than ballistic. Examples of such systems are the cyclohexane interconversion in a solvent [631] and the diffusion of nitrogen in an argon crystal [632].

N.23 Single particle in a two-dimensional potential well

Case Study 24 (Single particle in a two-dimensional potential well). To illustrate the path sampling method, consider a system containing a single particle in the following simple two-dimensional potential [641]:

$$V(x, y) = \left(4(1 - x^2 - y^2)^2 + 2(x^2 - 2)^2 + ((x + y)^2 - 1)^2 + ((x - y)^2 - 1)^2 - 2\right) / 6. \quad (\text{N.23.1})$$

Note that $V(x, y) = V(-x, y) = V(x, -y)$. Fig. N.36 shows that this potential consists of two stable regions around the points $(-1, 0)$, which we call A, and $(1, 0)$, which we call region B. To be more specific, all points within a distance of 0.7 from $(-1, 0)$ or $(1, 0)$ are defined to be in region A or B, respectively. At a temperature of $T = 0.1$ transitions from A to B are rare events.

To compute the rate of transitions from A to B we used path ensemble simulations. The initial distribution $\mathcal{N}(x_0)$ was chosen to be canonical, i.e.,

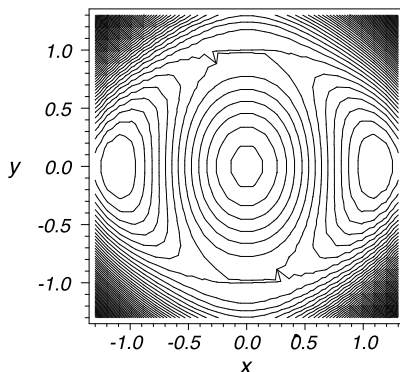


FIGURE N.36 Contour plot of the function $V(x, y)$ defined by Eq. (N.23.1). The two minima are at $(-1, 0)$, A, and $(0, 1)$, B. These minima are separated by a potential energy barrier.

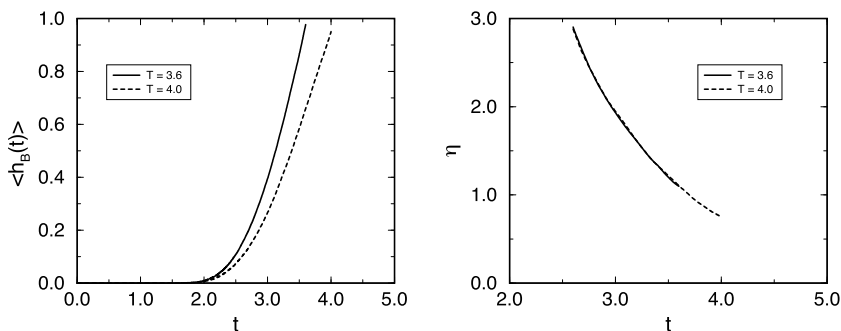


FIGURE N.37 $h_B(t)$ (left) and $\eta(t)$ (right) as a function of time for various values of the total path length T .

$$\mathcal{N}(x_0) \propto \exp[-\beta \mathcal{H}(x_0)].$$

A trajectory was generated using standard Molecular Dynamics simulations (see Chapter 4). The equations of motion were integrated using the velocity-Verlet algorithm with a time step of 0.002.

The first step was the calculation of the coefficient $\eta(t, t')$. This involves the computation of the path ensemble averages $\langle h_B(x_t) \rangle_{A, H_B(\mathcal{T})}$ for various times t . The result of such a simulation is shown in Fig. N.37 for $\mathcal{T} = 4.0$ and $\mathcal{T} = 3.6$. An important question is whether the time $\mathcal{T}$ is long enough. Since we are interested in the plateau of $k(t)$, the function $\langle h_B(x_t) \rangle_{A, H_B(\mathcal{T})}$ must have become a straight line for large values of t . If this function does not show a straight line, the value of $\mathcal{T}$ was probably too short, the process is not a rare event, or the process cannot be described by a single hopping rate. The consistency of the simulations can be tested by comparing the results with a simulation using a shorter (or longer, but this is more expensive) T . Fig. N.37 shows that the results of the two simulations are consistent.

The next step is the calculation of the correlation function $C(t)$. For the calculation of $P(\lambda, t)$, we have defined the order parameter λ as the distance from

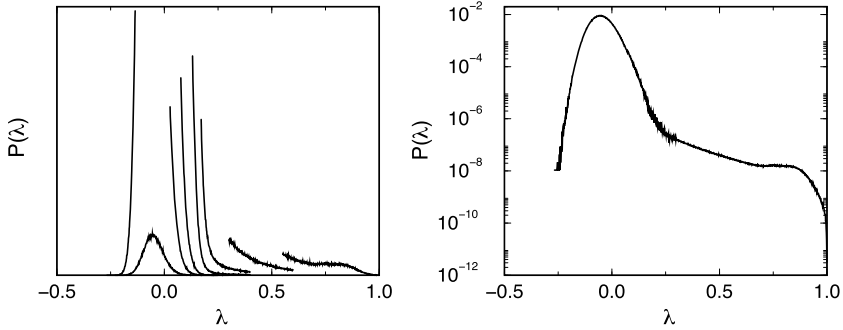


FIGURE N.38 (Left) $P(\lambda, i, t = 3.0)$ for all slices i . (Right) $P(\lambda, t = 3.0)$ when all slices i are combined. The units on the y axis are such that $\int_{-\infty}^1 d\lambda P(\lambda, t) = 1$.

point B:

$$\lambda = 1 - \frac{|\mathbf{r} - \mathbf{r}_B|}{|\mathbf{r}_A - \mathbf{r}_B|}, \quad (\text{N.23.2})$$

in which $\mathbf{r}_B = (1, 0)$. In this way, the region B is defined by $0.65 < \lambda \leq 1$, and the whole phase space is represented by $(-\infty, 1]$. In Fig. N.38 (left), we have plotted $P(\lambda, i, t = 3.0)$ as a function of λ for different slices i . Recombining the slices leads to Fig. N.38 (right). The value of $C(t = 3.0)$ can be obtained by integrating over region B:

$$C(t) = \int_B d\lambda P(\lambda, t). \quad (\text{N.23.3})$$

Combining the results gives for the total crossing rate

$$k = \eta(t) C(t). \quad (\text{N.23.4})$$

Using $t = 3.0$ leads to $\eta(2.0) = 1.94$, $C(3.0) = 4.0 \times 10^{-6}$, and $k = 8.0 \times 10^{-6}$.

N.24 Dissipative particle dynamics

Case Study 25 (Dissipative particle dynamics). To illustrate the [Dissipative Particle Dynamics \(DPD\)](#) technique, we have simulated a system of two components (1 and 2). The conservative force is a soft repulsive force, given by

$$\mathbf{f}_{ij}^C = \begin{cases} a_{ij} (1 - r_{ij}) \hat{\mathbf{r}}_{ij} & r_{ij} < r_c \\ 0 & r_{ij} \geq r_c \end{cases}, \quad (\text{N.24.1})$$

in which $r_{ij} = \|\mathbf{r}_{ij}\|$ and r_c is the cutoff radius of the potential. The random forces are given by Eq. (16.1.3) and the dissipative forces by Eq. (16.1.2). The total force on a particle equals the sum of the individual forces:

$$\mathbf{f}_i = \sum_{i \neq j} (\mathbf{f}_{ij}^C + \mathbf{f}_{ij}^S + \mathbf{f}_{ij}^R + \mathbf{f}_{ij}^D). \quad (\text{N.24.2})$$

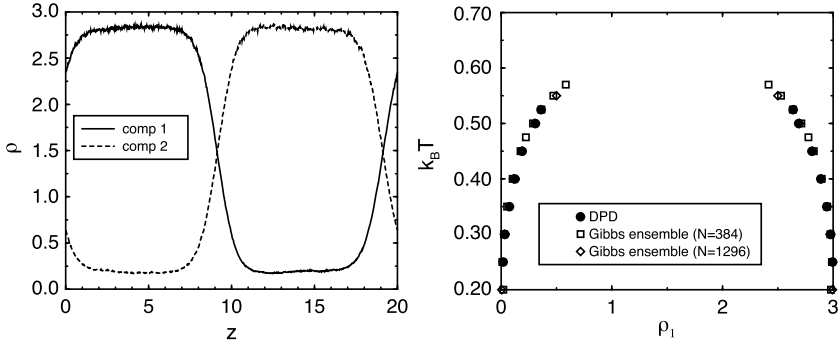


FIGURE N.39 (Left) Density profile for $k_B T = 0.45$. (Right) Phase diagram as calculated using DPD and Gibbs ensemble simulations. Both techniques result in the same phase diagram, but the Gibbs ensemble technique needs fewer particles due to the absence of a surface. In the DPD simulations, we have used a box of $10 \times 10 \times 20$ (in units of r_c^3). The time step of the integration was $\Delta t = 0.03$.

To obtain a canonical distribution, we use

$$\sigma^2 = 2\gamma k_B T$$

$$w^D(r_{ij}) = \left[w^R(r_{ij}) \right]^2.$$

A simple but useful choice is [682]

$$w^D(r_{ij}) = \begin{cases} (1 - r_{ij}/r_c)^2 & r_{ij} < r_c \\ 0 & r_{ij} \geq r_c \end{cases}$$

with $r_c = 1$. The simulations were performed with $\rho = 3.0$ and $\sigma = 1.5$. We have chosen $a_{ii} = a_{jj} = 25$ and $a_{ij, i \neq j} = 30$. This system will separate into two phases. In the example shown in Fig. N.39, we have chosen the z -direction perpendicular to the interface. The left part of Fig. N.39 shows typical density profiles of the two components. In Fig. N.39 (right), we have plotted the concentration of one of the components in the coexisting phases.

Since we can write down a Hamiltonian for a DPD system, we can also perform standard Monte Carlo simulations [691]. For example, we can also use a Gibbs ensemble simulation (see section 6.6) to compute the phase diagram. As expected, Fig. N.39 shows that both techniques give identical results. Of course, due to the presence of an interface one needs many more particles in such a DPD simulation.

Thermodynamic quantities are calculated using only the conservative force. The pressure of the system is calculated using

$$p = \rho k_B T + \frac{1}{3V} \sum_{i>j} \left(\mathbf{r}_{ij} \cdot \mathbf{f}_{ij}^C \right).$$

If the DPD fluid undergoes phase separation into two slabs containing the different fluid phases, and we can compute the associated interfacial tension using the

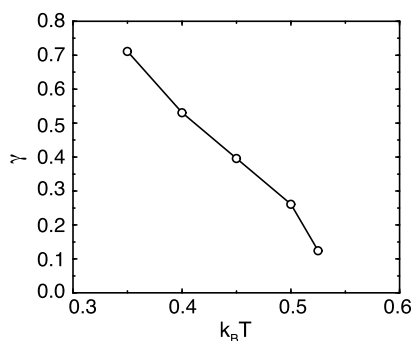


FIGURE N.40 Surface tension as a function of temperature.

techniques described in Section 5.1.6. Fig. N.40 shows this interfacial tension γ as a function of temperature. Clearly, γ decreases with increasing temperature and should vanish at the critical point. As the critical point is approached, the driving force for the formation of a surface (surface tension) becomes very low, and it becomes difficult, therefore, to form a stable interface in a small system.

N.25 Comparison of schemes for the Lennard-Jones fluid

Case Study 26 (Comparison of schemes for the Lennard-Jones fluid). It is instructive to make a detailed comparison of the various schemes to save CPU time for the Lennard-Jones fluid. We compare the following schemes:

1. Verlet list
2. Cell list
3. Combination of Verlet and cell lists
4. Simple N^2 algorithm

We have used the program of Case Study 1 as a starting point. At this point it is important to note that we have not tried to optimize the parameters (such as the Verlet radius) for the various methods; we have simply taken some reasonable values.

For the Verlet list (and for the combination of Verlet and cell lists) it is important that the maximum displacement be smaller than twice the difference between the Verlet radius and cutoff radius. For the cutoff radius we have used $r_c = 2.5\sigma$, and for the Verlet radius $r_v = 3.0\sigma$. This limits the maximum displacement to $\Delta_x = 0.25\sigma$ and implies for the Lennard-Jones fluid that, if we want to use a optimum acceptance of 50%, we can use the Verlet method only for densities larger than $\rho > 0.6\sigma^{-3}$. For smaller densities, the optimum displacement is larger than 0.25. Note that this density dependence does not exist in a Molecular Dynamics simulation. In a Molecular Dynamics simulation, the maximum displacement is determined by the integration scheme and therefore is independent of density. This makes the Verlet method much more appropriate for a Molecular Dynamics simulation than for a Monte Carlo simulation. Only at high densities does it make sense to use the Verlet list.

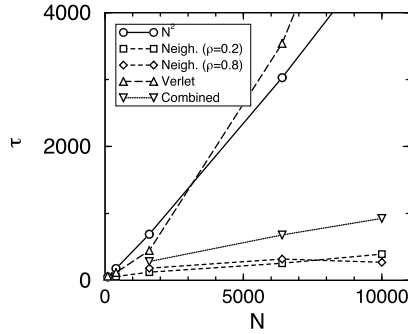


FIGURE N.41 Comparison of various schemes to calculate the energy: τ is in arbitrary units and N is the number of particles. As a test case, the Lennard-Jones fluid is used. The temperature was $T^* = 2$ and per cycle, the number of attempts to displace a particle was set to 100 for all systems. The lines serve to guide the eye.

The cell list method is advantageous only if the number of cells is larger than 3 in at least one direction. For the Lennard-Jones fluid this means that if the number of particles is 400, the density should be lower than $\rho < 0.5\sigma^{-3}$. An important advantage of the cell list over the Verlet list is that this list can also be used for moves in which a particle is given a random position.

From these arguments, it is clear that if the number of particles is smaller than 200–500, the simple N^2 algorithm is the best choice. If the number of particles is significantly larger and the density is low, the cell list method is probably more efficient. At high density, all methods can be efficient, and we have to make a detailed comparison.

To test these conclusions about the N dependence of the CPU time of the various methods, we have performed several simulations with a fixed number of Monte Carlo cycles. For the simple N^2 algorithm, the CPU time per attempt is

$$\tau_{N^2} = cN,$$

where c is the CPU time required to calculate one interaction. This implies that the total amount of CPU time is independent of the density. For a calculation of the total energy, we have to do this calculation N times, which gives the scaling of N^2 . Fig. N.41 shows that, indeed for the Lennard-Jones fluid, the τ_{N^2} increases linearly with the number of particles.

If we use the cell list, the CPU time will be

$$\tau_l = cV_l\rho + c_l p_l N,$$

where V_l is the total volume of the cells that contribute to the interaction (in three dimensions, $V_l = 27r_c^3$), c_l is the amount of CPU time required to make a cell list, and p_l is the probability that a new list has to be made. Fig. N.41 shows that the use of a cell list reduces the CPU time for 10,000 particles with a factor 18. Interestingly, the CPU time does not increase with increasing density. We would expect an increase since the number of particles that contribute to the interaction of a particle i increases with density. However, the second contribution to τ_{Neigh}

(p_l) is the probability that a new list has to be made, depends on the maximum displacement, which decreases when the density increases. Therefore, this last term will contribute less at higher densities.

For the Verlet scheme, the CPU time is

$$\tau_v = cV_v\rho + c_v p_v N^2,$$

where V_v is the volume of the Verlet sphere (in three dimensions, $V_v = 4\pi r_v^3/3$), c_v is the amount of CPU time required to make the Verlet-list, and p_v is the probability that a new list has to be made. Fig. N.41 shows that this scheme is not very efficient. The N^2 operation dominates the calculation. Note that we use a program in which a new list for all particles has to be made as soon as one of the particles has moved more than $(r_v - r_c)/2$; with some more bookkeeping, it is possible to make a much more efficient program, in which a new list is made for only the particle that has moved out of the list.

The combination of the cell and Verlet lists removes the N^2 dependence on the simple Verlet algorithm. The CPU time is given by

$$\tau_c = cV_v\rho + c_v p_v c_l N.$$

Fig. N.41 shows that indeed the N^2 dependence is removed, but the resulting scheme is not more efficient than the cell list alone.

This case study demonstrates that it is not simple to give a general recipe for which method to use. Depending on the conditions and number of particles, different algorithms are optimal. It is important to note that for a Molecular Dynamics simulation, the conclusions may be different.